

BIOLOGY GRADE 11: REPRODUCTION IN PLANTS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.1 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	11
Subject	Biology
Strand	Strand 2.0: Anatomy and Physiology of Plants
Sub-Strand	Sub-Strand 2.1: Reproduction in Plants
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Asexual reproduction (sporulation and vegetative propagation, including specialised vegetative structures such as runners, bulbs, corms, rhizomes, offshoots, tubers and cuttings) – Sexual reproduction in flowering plants: identification of flowers based on the position of the ovary, symmetry, inflorescence, and presence of male and female parts – Mechanisms and features that hinder self-pollination and self-fertilisation – Double fertilisation and its significance – Seed and fruit development (including placentation and classification of fruits)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - describe asexual reproduction in plants, - identify flowers based on their floral parts, - illustrate mechanisms and features that hinder self-pollination and self-fertilisation in plants, - describe the process of double fertilisation in flowering plants, - acknowledge the importance of sexual and asexual reproduction in plants.
Core Competencies	– Communication and Collaboration: The learner cooperates with others while observing and classifying specimens of flowers from different plants (Lessons 1 and 2). – Critical Thinking and Problem Solving: The learner analyses causes and effects while brainstorming on the advantages and disadvantages of sexual and asexual reproduction in plants (Lesson 6).
Core Values	– Social Justice: The learner shares resources equitably while examining, drawing, and labelling sections of fruits (Lesson 5), e.g. sharing scalpels, hand lenses and fruit specimens within groups of 4-5. – Love: The learner prioritises others' interests while using models and charts to illustrate mechanisms and features that hinder self-pollination and self-fertilisation in plants (Lesson 3).
Science & Engineering Practices	– Asking Questions: Students generate questions about why cassava propagates from cuttings but maize needs seeds, and post them on the Driving Question Board (Lessons 1, 7). – Developing and Using Models: Students build and revise a cumulative model linking vegetative structures, floral adaptations and double fertilisation, using wire/bead flower models in Lesson 3 (Lessons 3, 7). – Planning and Carrying Out Investigations: Students dissect flowers and fruits using shared equipment and hand lenses to

	gather structural evidence (Lessons 2, 5). – Analysing and Interpreting Data: Students classify specimens by ovary position, symmetry and placentation and compare the patterns (Lessons 2, 5). – Engaging in Argument from Evidence: Students debate the trade-offs of sexual and asexual reproduction using case cards, e.g. banana wilt (Lesson 6). – Constructing Explanations: Students write Claim-Evidence-Reasoning explanations of the anchoring phenomenon (Lessons 7, 8).
Pertinent & Contemporary Issues (PCIs)	– Environment and Technology: The learner appreciates the use of technology in managing the environment while watching animations illustrating double fertilisation and formation of seeds and fruits (Lesson 4). – Online Safety: The learner explores safe internet sites while watching animations illustrating double fertilisation and formation of seeds and fruits (Lesson 4). Animations are pre-downloaded/cached by the teacher given limited internet.
Career Connections	– Agronomist: applies knowledge of propagation and pollination to improve crop yields on Kenyan farms. – Plant Breeder: uses cross-pollination and genetic variation to develop improved maize, bean and avocado varieties. – Agricultural Extension Officer: advises farmers in areas such as Kericho, Kisumu and the Rift Valley on cuttings, planting pollinator trees and seed selection. – Botanist: classifies flowers and fruits by structure and studies reproductive adaptations. – Horticulturalist: manages vegetative propagation and fruit set in orchards of pawpaw, avocado and passion fruit. – Crop Geneticist: studies genetic variation and disease resistance, e.g. banana wilt and cassava varieties.
Focus for Lessons	This unit uses a local farming puzzle (cuttings that grow without seeds versus fruit trees that need a partner tree) to drive investigation of asexual and sexual reproduction in plants. Students observe local specimens, dissect flowers and fruits, model pollination barriers, study double fertilisation through animations, and build and revise a cumulative explanatory model. The unit ends with Claim-Evidence-Reasoning explanations and assessment, designed for large classes with shared equipment and cached digital resources.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do some farm plants produce offspring without seeds while others require seeds and a partner tree to make fruit? KICD KEY INQUIRY QUESTIONS: 1. What is the significance of reproduction in plants? 2. How does reproduction occur in plants?
Anchoring Phenomenon	Farmers in the community plant cassava, sugarcane and sweet potato cuttings and get new plants without any seeds, yet maize and beans must be grown from seeds. Furthermore, a single pawpaw or avocado tree in a compound sometimes flowers heavily but never sets fruit unless another tree grows nearby. Students eat these crops daily (cassava, sweet potatoes, passion fruit, avocado, ugali from maize) but rarely wonder why they are propagated and fruited so differently. It is puzzling because a lone tree can be covered in flowers and still produce nothing, while a piece of stem can become a whole new plant with no flower at all.
Storyline Thread	Lesson 1: Plants can reproduce asexually through sporulation and specialized structures without forming seeds. Students observe and classify local specimens (sweet potato tubers, onion bulbs, ginger rhizomes, grass runners, bryophyllum leaves) using hand lenses and a chart. Leads to: How do plants reproduce sexually through flowers, and do all flowers have the same structures? Lesson 2: Flowers vary by ovary position (superior/inferior), symmetry, inflorescence, and unisexual/bisexual parts. Students examine and dissect real flowers (Hibiscus, Bougainvillea, maize inflorescence, passion flower) and classify them on a worksheet. Leads to: If flowers have both male and female parts, why don't all flowers pollinate themselves?

	Lesson 3: Specific adaptations (protandry, protogyny, heterostyly, dioecy, self-incompatibility) prevent self-pollination. Students use models and structural charts to trace pollen movement and simulate structural hindrances. Leads to: What actually happens inside the flower after pollen successfully lands on a receptive stigma? Lesson 4: Double fertilisation occurs when one sperm nucleus fuses with the egg cell (zygote) and another fuses with polar nuclei (endosperm). Students watch cached animations of pollen tube growth and diagram the pathway. Leads to: How does double fertilisation transform the ovary and ovules into fruits and seeds? Lesson 5: Fruits and seeds develop post-fertilisation, displaying distinct placentation types (marginal, axile, parietal, free-central). Students section, examine, draw and label local fruits (orange, bean pod, tomato, passion fruit, cucumber) to classify placentation. Leads to: Is sexual reproduction better than asexual reproduction for a plant's survival and for farming? Lesson 6: Sexual reproduction creates genetic variation, while asexual reproduction allows rapid colonization of identical traits. Students brainstorm and debate trade-offs using a comparative chart and case cards (e.g. banana wilt vs. seed variation). Leads to: How can we synthesize our findings to explain our local crop phenomenon completely? Lesson 7: Integration of evidence: how seedless propagation vs. cross-pollinated fruiting operates in local agriculture. Students revise their Day 1 models and present group explanatory models connecting vegetative structures, floral adaptations and double fertilisation, using the Driving Question Board. Leads to: How do we format our complete scientific explanation for final assessment? Lesson 8: Final synthesis, peer review, and assessment of plant reproduction mechanisms. Students finalise individual Claim-Evidence-Reasoning explanations and complete the unit assessment tasks using the rubric. Unit completion.
Supporting Phenomena	– Bryophyllum (leaf-of-life) leaves drop on soil and grow tiny plantlets along their edges with no seeds or flowers. – Ginger rhizomes, onion bulbs and sweet potato tubers sprout new shoots in the kitchen or market stall. – Maize plants have tassels (male) and cobs with silks (female) on the same plant, yet cobs fill with kernels only when pollen reaches the silks. – A banana plantation propagated from suckers can be wiped out by banana wilt, while a field of beans grown from seed shows varied survival.

ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (80 minutes (double lesson: 2 x 40 minutes)): Cuttings, Tubers and Lonely Trees: Why Do Plants Reproduce So Differently?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon (cassava, sugarcane and sweet potato grow from cuttings without seeds, while maize and beans need seeds, and a lone pawpaw or avocado tree flowers heavily but sets no fruit) and to open the Driving Question Board. Learners gather first evidence that plants can reproduce asexually, without seeds, through sporulation and specialised structures.
Knowledge	By the end of the lesson the learner should be able to: (a) describe asexual reproduction in plants (sporulation and vegetative propagation through runners, bulbs, corms, rhizomes, offshoots, tubers and cuttings); (b) identify and classify local specimens (sweet potato tubers, onion bulbs, ginger rhizomes, grass runners, bryophyllum leaves) by the structure that produces new plants; (c) begin to acknowledge the importance of asexual reproduction in plants.
Skills	NGSS Science and Engineering Practices: Developing and Using Models (initial model of how a sweet potato tuber makes a new vine versus how a bean seed makes a plant); Asking Questions (DQB sticky notes); Analysing and Interpreting Data (classifying specimens using a chart and a hand lens); Constructing Explanations. Core competencies: Communication and Collaboration (cooperating with others while observing and classifying specimens); Critical Thinking and Problem Solving (analysing causes and effects of seedless propagation).
Attitudes	Values: Social Justice (sharing specimens and hand lenses equitably in groups of 4-5); Love (prioritising others' interests and ideas during group discussion). PCI: Environment and Technology (appreciating how knowledge of propagation helps farmers manage crops and the environment). Curiosity and respect for local farmers' knowledge.
Key Inquiry Question	1. What is the significance of reproduction in plants? 2. How does reproduction occur in plants?
Purpose in Storyline	Lesson 1 sets the puzzle and the DQB with two columns, 'Without Seeds' and 'With Seeds'. Learners discover that many familiar Kenyan crops multiply without seeds through specialised structures. This raises the question for Lesson 2: How do plants reproduce sexually through flowers, and do all flowers have the same structures? It also starts to explain why the lone pawpaw or avocado flowers but gives no fruit (the flowers need something that vegetative structures do not).
Safety Notes	No blades are used in this lesson. Learners handle specimens only with hands and hand lenses. Wash hands after handling soil-covered tubers, bulbs and rhizomes. Do not taste any specimens (the bryophyllum leaf is not for eating). Warn about allergic reactions: onion bulbs may irritate the eyes, so do not rub eyes. Keep hand lenses away from direct sunlight (never look at the sun through them). Carry bags to the bin, and keep walkways clear in the crowded room. Safety protocols for blades and razors will be taught explicitly in Lessons 2 and 5.

B. LESSON OVERVIEW

The teacher opens with a Kenyan farm scene: a farmer in Kisii or Kakamega cutting cassava stems and planting them, a sugarcane farmer in Kisumu planting cane pieces, and a woman in Siaya planting sweet potato vines, while the same farmers buy maize and bean seed before the long rains. A home compound in Nairobi has a pawpaw or avocado tree covered in flowers that never fruits unless another tree is nearby. Learners predict and draw an initial model comparing a sweet potato tuber making a new vine with a bean seed making a plant. In groups of 4-5 they observe and classify sweet potato tubers, onion bulbs, ginger rhizomes, grass runners and bryophyllum leaves with hand lenses and a chart on asexual structures. They explain how each structure gives rise to new plants without seeds, post questions on the DQB under 'Without Seeds' and 'With Seeds', and revise their first model. The lesson closes with the question that drives Lesson 2: how do plants reproduce sexually through flowers, and do all flowers have the same structures?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Plant reproductive success Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Asexual Reproduction Practice Source: CK-12  Search ARES for similar readings	Learners view a chart or teacher sketch (or a short pre-downloaded clip or photos) showing: (1) a farmer planting cassava stem cuttings, sugarcane pieces and sweet potato vines; (2) a farmer planting maize and bean seeds; (3) a lone avocado or pawpaw tree full of flowers but with no fruit. Individually, for 5 minutes, each learner draws an initial model on a half page of their exercise book showing how a sweet potato tuber or vine creates a new plant, compared with how a bean seed creates a plant. Models must include labels and arrows and a short caption: 'I think the new plant comes from...'. They then add a prediction: 'The lone avocado has flowers but no fruit because...'. They share their drawings with a shoulder partner.	Before class, display the phenomenon pictures (or ask 3 learners to bring cassava cutting, ugali flour, and avocado photo). Say: 'Look at what we eat every day: ugali from maize, boiled cassava, sweet potatoes, avocado. Notice how farmers in our community plant them.' Pose: 'Why can a piece of cassava stem become a whole plant with no flower and no seed, but maize and beans must be planted as seeds? And why does a lone avocado tree full of flowers give no fruit?' Give 10-15 seconds of WAIT TIME with no answers taken. Say: 'There are no wrong ideas today. I want your first thinking.' Circulate during drawing and note 5 common ideas (e.g., 'the tuber is a seed', 'roots come from the flower'). Cold-call 4 learners from different parts of the room to share one thing in their model, and ask 'What made you draw it that way?' Record the ideas in a column on the board without evaluating. Keep the drawings, since learners will revise them in Phase 5.	Predict-Observe-Explain (POE) - the Predict step. Learners commit to a model before seeing evidence, which exposes preconceptions (for example that all plants need seeds or that a flower alone guarantees fruit). Later evidence is compared directly against the prediction, so conceptual change is visible to learners and to the teacher.	Observable indicators: (1) each learner's model shows at least two labelled parts (tuber/vine/roots or seed/shoot/root); (2) the model shows a clear starting point and an arrow showing growth; (3) learner writes a prediction sentence about the lone fruit tree. Teacher tallies with a quick check-mark list: how many learners draw a seed inside the sweet potato, and how many draw only a flower. Use the tally to plan Phase 3 questions. Keep drawings dated for comparison later.
Observe Phase  VIDEO: Sexual and asexual reproduction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	In groups of 4-5 (about 10 groups; sharing one specimen tray and 2 to 3 hand lenses per group), learners examine five specimens: sweet potato tuber (with sprouting buds or 'eyes' if available), onion bulb (cut open by the teacher beforehand to show fleshy leaves and a stem plate), ginger rhizome (with nodes and buds), grass runner (for example Kikuyu grass	Before class, prepare 10 trays with the five specimens and labelled numbers; cut the onion bulb vertically in advance (the teacher handles any cutting, no blades for learners). Say: 'Today you are plant detectives. Your job is to find where a new plant could start on each specimen.' Model one example at the front: hold the ginger rhizome and ask 'What do	Predict-Observe-Explain (POE) - the Observe step, combined with classification. Sorting specimens by structure forces learners to notice shared features (buds or nodes, stored food, no flower or seed involved), so they can build the general idea that these parts act as 'starter kits' for new plants.	Observable indicators: (1) group table has all five specimens correctly matched to a structure type (check at least 4 of 5 correct); (2) evidence column cites something seen (for example 'nodes with tiny buds'), not just a name; (3) every learner handles a specimen and a hand lens (teacher observation checklist); (4) learners correctly say that spores

 EXERCISE: Understand: sexual and asexual reproduction Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	or couch grass stolon with nodes and roots), and bryophyllum leaf with plantlets on the margin. Using the chart on asexual structures and a hand lens, each group completes a table with columns: Specimen, Structure type (tuber, bulb, rhizome, runner, leaf plantlet), Where the new plant would grow from (buds, nodes, margin), Food stored? (yes/no), Evidence (what I saw). Roles are rotated: Observer, Recorder, Materials Manager, Reporter. Learners also read a short note: 'Cassava, sugarcane and sweet potato are planted as cuttings, offshoots are used in bananas, and ferns and mushrooms use spores (sporulation).' Groups then add a column marking each specimen's structure and one Kenyan crop or plant that uses it (for example bananas and suckers, Irish potatoes in Nyandarua as tubers, onions in Kajiado bulbs, arrowroots as rhizomes, kikuyu grass as runners, strawberries as runners).	you see that looks like a place for a new shoot?' WAIT TIME 10-15 seconds, then point out the nodes and buds. Remind safety: 'Wash hands after touching soil, do not rub your eyes after handling onion, hand lenses never point at the sun, and no tasting.' Circulate and ask each group two probing questions: 'Which part of this will become the new shoot? How do you know?' and 'Where is the stored food, and why would a new plant need it?' Cold-call 3 groups to hold up a specimen and name the structure type. Prompt equity: 'Make sure everyone in your group gets a turn with the hand lens; the Materials Manager must pass it every 2 minutes.' Mention that mushrooms, bread mould and ferns reproduce by spores (sporulation): show a picture of bread mould or a fern frond with sori, and ask 'How is a spore different from a seed?' Take 2 responses.		differ from seeds in at least one way (for example spores are single cells and have no stored food, while a seed contains an embryo and food store). Teacher uses a clipboard to mark groups needing support.
Explain Phase  VIDEO: Plant reproductive success Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Asexual vs. Sexual Reproduction Practice Source: CK-12  Search ARES for similar readings	Learners return to their desks and use their table to explain: 'How can a piece of sweet potato vine or a cassava stem become a whole new plant without a seed?' They write a three-part Claim-Evidence-Reasoning (CER) statement: Claim (for example 'Sweet potato, ginger, onion and Kikuyu grass can make new plants without seeds'), Evidence (what they observed on the specimens), Reasoning (the new plant grows from buds or nodes by cell division and uses stored food, so the offspring is genetically identical to the parent because only one parent is involved). Then in Think-	Say: 'Use your table. A good explanation uses what you saw, not what you guessed.' Display the CER sentence frames on the board: 'I claim that... My evidence is... This shows that... because...'. Give 3 minutes of silent writing, then pair talk for 2 minutes, then a share. Use WAIT TIME of 10-15 seconds after each question. Ask: 'If every cassava plant on the farm grows from cuttings of the same parent, how alike will they be?' Cold-call 4 pairs (not volunteers) and press for reasoning: 'What in your evidence supports that?' Build the idea that asexual reproduction is fast, needs one parent, and	Claim-Evidence-Reasoning (CER) paired with Think-Pair-Share. CER requires learners to connect their classification table to a general principle rather than just recalling names. Think-Pair-Share gives every one of the 45-50 learners a voice in a large class, and discussing advantages and disadvantages develops Critical Thinking and Problem Solving.	Observable indicators: (1) CER statement has a claim that mentions at least two structures; (2) evidence refers to specific features observed (buds, nodes, stored food, plantlets); (3) reasoning mentions that no seed or second parent is needed or that offspring are identical to the parent; (4) learners list at least one advantage and one disadvantage. Collect 10 CER statements at random as a quick sample and sort them into 'complete', 'partial' and 'not yet'. Use these to plan the opening of Lesson 2.

	Pair-Share they discuss: 'What are the advantages for a farmer of growing cassava from cuttings rather than seeds? What could be a disadvantage if a disease attacks the farm?' Pairs share with another pair and then the class. Learners also answer: 'Does this explain why the lone avocado has flowers but no fruit?' and decide 'not yet'.	produces identical offspring; also connect to risk: 'If one cassava plant is weak to cassava mosaic disease, what might happen to the others?' Take 2 responses. Then pose the tension: 'So why do maize and beans use seeds? And why does the lone avocado not fruit?' Deliberately leave this unanswered and say: 'Those are questions for our board.' Address misconceptions: a tuber is not a seed; a bulb is a stem with fleshy leaves.		
Driving Question Board (DQB) Creation  VIDEO: Plant reproductive success Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Asexual Reproduction Practice Source: CK-12  Search ARES for similar readings	The class opens the DQB. The teacher has drawn two columns on a manila paper or a section of the chalkboard: 'Without Seeds' and 'With Seeds', with the driving question at the top: 'Why do some farm plants produce offspring without seeds while others require seeds and a partner tree to make fruit?' Each learner writes one or two questions on sticky notes (or small paper slips with tape if sticky notes are not available), for example 'Why does maize need seed?', 'Why does a lone pawpaw not give fruit?', 'Can bean plants grow from cuttings?', 'What is inside a seed?', 'What do the flowers do?'. Learners post their notes in the correct column and then, in groups, cluster similar questions and suggest a heading for each cluster (for example 'Flowers and fruits', 'Partner trees', 'Seed structure'). Questions from Lesson 1 evidence such as 'Why do onions make bulbs?' go under 'Without Seeds'.	Say: 'Great scientists begin with great questions. What are you still wondering after today?' Give 2 minutes for silent writing with WAIT TIME 10-15 seconds before taking any questions. Ask each group to send two members (cold-call 10 groups, 20 learners) to post notes in the right column. Read out 3 questions and ask the class 'Which column does this belong in, and why?' Model good question stems: 'Why...', 'How...', 'What would happen if...'. Highlight any question about the lone fruit tree and say: 'Many of you ask about flowers. That is where we go next.' Encourage open questions rather than yes/no. Because boards are shared by large classes, assign two learners as 'DQB keepers' to tidy and photograph the board each week. Reinforce the value of Love by asking learners to read and value classmates' questions: 'Find a question from another group that you also want answered.'	Driving Question Board (DQB). The DQB turns learners' own wonderings into the agenda for the unit, shows how each lesson answers some questions and raises new ones, and gives ownership of the learning. Sorting into 'Without Seeds' and 'With Seeds' helps learners see the structure of the problem and build categories.	Observable indicators: (1) every learner contributes at least one question; (2) at least 70% of notes are 'why' or 'how' questions rather than simple fact questions; (3) learners place notes in the correct column and can justify the placement; (4) at least 5 questions refer to flowers, pollination, partner trees or fruit formation, showing readiness for Lesson 2. Teacher records the number of questions per cluster and photographs the DQB for later reference.
Model Building Phase  VIDEO:	Learners return to their Phase 1 model of how a sweet potato tuber makes a new vine compared with how a bean seed makes a plant.	Say: 'Take out your first model. What would you change now that you have seen the evidence? Use a different colour so we can see	Cumulative model revision. Revising a personal model in a different colour makes learning visible, shows conceptual change	Observable indicators: (1) revised model includes buds or nodes and stored food as the source of the new plant; (2) revised model states

Plant reproductive success Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Asexual Reproduction Practice Source: CK-12 Search ARES for similar readings	Using a different coloured pen, they revise their drawing: add the buds or nodes on the tuber or rhizome, show stored food, show roots and shoots developing from the bud, label 'no seed, one parent', and add a small box for 'questions I still have' (for example 'What does a seed have that a tuber does not?'). They add the five specimens as a mini key under the model. As an exit ticket, each learner answers on a slip: (1) Name two plant structures used for asexual reproduction and one Kenyan plant for each. (2) One advantage of reproducing without seeds. (3) One question still unanswered about the lone avocado tree.	your thinking grow.' Give 3 minutes of silent revising with WAIT TIME of 10-15 seconds after the instruction before you begin to circulate. Use oral probes at 6 desks: 'What did you add? What evidence made you add it?' and 'What part of your model still does not explain the lone avocado?' Cold-call 3 learners to share one change. Announce the next lesson: 'Cassava and sweet potato need no flower. But the avocado needs flowers and a partner tree. Next time we ask: How do plants reproduce sexually through flowers, and do all flowers have the same structures?' Ask learners to bring any flowers they can find safely at home (hibiscus, bean, tomato, passion fruit, maize tassel) wrapped in a damp cloth. Collect exit tickets at the door and sort them into three piles: ready, nearly, and needs support.	against the first prediction (the Explain step of POE), and records new questions that carry forward the storyline into sexual reproduction in flowers.	or implies one parent and no seed; (3) the exit ticket names two correct structures with a local example each (target: 80% of learners); (4) at least one genuine question is written about the lone fruit tree. Sort exit tickets and record the number with misconceptions (for example 'a tuber is a seed') for reteaching at the start of Lesson 2.
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D. TEACHER REFLECTION

After the lesson ask yourself: 1. Did all learners, including those at the back of a class of 45-50, handle a specimen and use a hand lens? 2. Which misconceptions appeared in the first models (for example 'all plants need seeds', 'a tuber is a seed', 'flowers always give fruit') and did the revised models show change? 3. Were the DQB questions mostly 'why' and 'how' questions, and did enough point toward flowers and pollination to launch Lesson 2? 4. Were the specimens available and fresh (sweet potato, onion, ginger, grass runner, bryophyllum), and what local substitutes (arrowroot, banana sucker, Irish potato, couch grass) could be used next time? 5. Was WAIT TIME actually kept at 10-15 seconds, and did cold-calling reach quiet learners? 6. What changes are needed for the safety talk and equipment sharing before the blade-handling in Lessons 2 and 5? Note the learners who need extra support and prepare to group them with strong partners.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did you observe on the sweet potato tuber, onion bulb, ginger rhizome, grass runner and bryophyllum leaf? Record the structure and the place where a new plant could start (buds, nodes, leaf margin).
What did I learn?	What did you learn about how plants such as cassava, sugarcane, sweet potato, ginger and onion can reproduce without seeds? Include the idea of one parent, stored food and identical offspring, and name at least one advantage and one disadvantage.
How does this explain the phenomenon?	How does this evidence help to explain part of the driving question, and what is still unexplained? Why might maize and beans need seeds, and why might a lone avocado or pawpaw tree full of flowers not set fruit? Write the new question that leads into

	Lesson 2.
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LESSON 2 (80 minutes (double lesson)): Flower Detectives: Classifying Flowers by Ovary Position, Symmetry, Inflorescence and Sex

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners examine and dissect real flowers to identify floral parts and classify flowers by ovary position, symmetry, inflorescence type and presence of male and female parts, so they can explain why a lone pawpaw or avocado tree may flower heavily yet fail to set fruit.
Knowledge	Specific Learning Outcome (b): identify flowers based on their floral parts. Flowers vary by ovary position (superior/inferior), symmetry (actinomorphic/zygomorphic), inflorescence (single or clustered) and presence of male (stamens) and female (carpels) parts (bisexual or unisexual).
Skills	NGSS Science and Engineering Practices: Planning and Carrying Out Investigations (dissection and observation), Analyzing and Interpreting Data (classification table), Constructing Explanations (CER), Developing and Using Models (DQB model revision). Communication and Collaboration: the learner cooperates with others while observing and classifying specimens of flowers from different plants. Critical Thinking and Problem Solving: the learner analyses causes and effects.
Attitudes	Values: Social Justice, the learner shares resources equitably (blades, hand lenses, specimens) while examining, drawing and labelling. Love, the learner prioritises others' interests while working in groups. PCIs: Environment and Technology, appreciation of technology when using cached digital resources. Online Safety, using only the teacher-approved cached files.
Key Inquiry Question	1. What is the significance of reproduction in plants? 2. How does reproduction occur in plants?
Purpose in Storyline	Lesson 1 showed that cassava, sugarcane and sweet potato regrow from cuttings without seeds, while maize and beans need seeds. This lesson turns to the flower, the organ for seed-based reproduction. Learners gather structural evidence on which flowers have male parts, female parts or both. This sets up the puzzle: if flowers have both male and female parts, why don't all flowers pollinate themselves? That question leads into Lesson 3, on mechanisms that hinder self-pollination.
Safety Notes	RAZOR BLADE/SCALPEL PROTOCOL (explicit for Lessons 2 and 5): 1) Only one designated Cutter per group of 4-5 handles the blade, and the role rotates only when the blade is placed down in the tray. 2) Blades are kept in a labelled tray or blade-holder or mounted in a handle, never passed hand to hand; place the blade on the tray and let the next person pick it up. 3) Always cut away from the body and fingers, on a tile or cardboard board, with the flower held by forceps or a pencil and not in the palm. 4) Learners stand or sit upright and keep the bench clear of bags and food. 5) Teacher counts blades out and in at the start and end (record the number on the board). 6) Any cut: stop, raise hand, teacher applies first aid from the kit. 7) Dispose of used blades in a closed container (an old tin or bottle with a slit) and never in open bins. 8) Wash hands after handling plants, since Bougainvillea bracts and some plant saps can irritate skin. Do not taste any specimen. Wear a lab coat or apron if available. Digital files are pre-downloaded; no live internet use.

B. LESSON OVERVIEW

Learners begin by predicting what is inside a Hibiscus flower, then dissect Hibiscus, Bougainvillea, maize inflorescence and passion flower in groups of 4-5 sharing blades and hand lenses under a strict safety protocol. They record structures on a floral classification worksheet (ovary position, symmetry, inflorescence, bisexual/unisexual). They use Think-Pair-Share on placentation and Claim-Evidence-Reasoning to link structure to the phenomenon of lone pawpaw and avocado trees that flower but do not fruit. They add questions to the Driving Question Board, revise the class model, and leave with a new puzzle: if flowers have both male and female parts, why don't all flowers pollinate themselves? Time plan: Predict 12 min, Observe 28 min, Explain 20 min, DQB 8 min, Model 12 min.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: reproduction plants superior ovary  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: reproduction plants superior ovary	Learners recall the phenomenon: cassava and sweet potato cuttings grow with no flower, but a lone pawpaw or avocado tree can be covered in flowers and never set fruit. Individually, in their notebooks, they sketch a Hibiscus flower from memory and label the parts they think are inside, then predict: (a) Which flowers have both male and female parts? (b) Where is the ovary located in relation to the petals? (c) Why might a flower fail to make fruit? In groups they compare predictions and record one group prediction on the POE sheet, under the headings Predict, Observe and Explain.	Hold up a fresh Hibiscus flower and say: 'Remember the lone avocado tree covered in flowers that never gave fruit. What must be inside these flowers that decides whether fruit forms?' Give 2 minutes of silent individual sketching, then WAIT TIME of 10-15 seconds after the question 'Which flower on our table do you think has BOTH male and female parts, and why?' Cold-call 4 learners from different groups and write the predictions on the board without judging. Ask: 'Who disagrees, and what is your reasoning?' Distribute blades, hand lenses and worksheets by group leader (equitable sharing: check that each group has at least 1 hand lens per 2 learners, or rotate). Deliver the safety briefing (2 minutes) using the protocol, demonstrate the correct cut and blade placement on the tray, and ask 3 cold-called learners to repeat one rule each. Do not begin dissection until each group's Cutter has been named and has stated the rules.	Predict-Observe-Explain (POE): committing to a prediction before dissection exposes prior ideas (for example 'all flowers have the same parts') so that the evidence in the Observe phase can challenge or confirm them. This builds the habit of using structure as evidence for the phenomenon.	Check that 80% of learners have a labelled prediction sketch before moving on. Observable indicators: sketches show petals, stamens and a pistil or ovary; predictions give a reason and not just a guess. Note common ideas for later (for example 'every flower is male and female'). Safety check: every Cutter correctly states at least 3 of the 8 protocol rules.
Observe Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: reproduction plants superior ovary	Groups of 4-5 rotate through four specimen stations or work through four specimens in sequence: Hibiscus (large, bisexual, regular), Bougainvillea (small flowers surrounded by showy colourful bracts, clustered), maize inflorescence (male tassel and female cob silks on the same plant), and passion flower (complex, with a corona and stalked ovary). Roles: Cutter, Holder with forceps, Recorder,	Circulate with a checklist and the blade count. Say: 'Cutter, show me where your other hand is.' and 'Blade down on the tray before you pass anything.' Stop the whole class briefly at 8 minutes: 'Freeze. Look at your Hibiscus. Is the ovary above or below where the petals join? Point to it.' WAIT TIME 10-15 seconds, then cold-call 3 groups. At the maize station ask: 'Where are the male parts and where are the female parts? What do you	POE (Observe step): learners compare what they see with their written prediction, so cognitive conflict (for example Bougainvillea 'petals' are bracts; maize has separate male and female flowers) drives conceptual change. Collaborative role rotation keeps all 45-50 learners engaged despite shared equipment.	Labelled dissection diagram check: at least 6 correct labels (petal, sepal, stamen/anther, filament, stigma, style, ovary) for Hibiscus. Worksheet check: each specimen row has ovary position, symmetry, inflorescence and bisexual/unisexual completed with a supporting observation. Peer review: groups swap worksheets and use a 3-point rubric (accuracy, evidence noted, neatness) and give one star and one wish. Safety

 READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: reproduction plants superior ovary	Drawer/Reporter. Using hand lenses, learners: 1) count petals and sepals and decide symmetry (cut the flower in two along different lines: actinomorphic if more than one line gives equal halves, zygomorphic if only one); 2) carefully slice the flower lengthwise to see whether the ovary sits above (superior) or below (inferior) the point where petals attach; 3) find stamens (anthers and filaments) and carpel (stigma, style, ovary) and decide whether the flower is bisexual or unisexual; 4) note whether flowers occur singly or in a cluster (inflorescence: single, raceme, spike, cyme or tassel and cob); 5) cut across the ovary to see the ovules and how they are attached (placentation, the first clue for later). They complete the floral classification worksheet and draw one labelled diagram of the Hibiscus section. Tomato, bean or pawpaw flowers, if available, are added as extension specimens. Leftover or dried flowers, plus the teacher's cached photographs, cover specimens that are not in season.	notice about where they are on the plant?' (tassel versus silk; unisexual flowers on the same plant, monoecious). At Bougainvillea ask: 'Which parts are the real flowers and which are the colourful bracts? How can you tell?' For passion flower: 'How many stamens and stigmas? Which would an insect touch first?' Prompt equitable roles: 'Swap roles after each specimen so everyone gets to hold the hand lens.' Pose the key challenge: 'Does the evidence match your prediction? Record what surprised you.' Collect used blades at the end, count them in aloud with the class, and place them in the sharps container.		indicator: no cut injuries, blades stay on trays, and the count at the end matches the count at the start.
Explain Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: reproduction plants superior ovary  READING: No closely matching reading	Part A (10 min), Think-Pair-Share on placentation: learners look at the cross-section of the ovary they cut, count the chambers (locules), and see how the ovules are attached. Individually they think and sketch for 1-2 minutes, discuss with a partner for 2 minutes, then share with the class. Part B (10 min), CER on classification and the phenomenon: each pair writes a Claim-Evidence-Reasoning statement answering: 'Why might a lone pawpaw or avocado tree have	For Think-Pair-Share say: 'Look inside your ovary. How many chambers do you see, and where are the tiny ovules attached? Think quietly first.' WAIT TIME 10-15 seconds, pairs talk for 2 minutes, then cold-call 4 pairs for different specimens. Record on the board: 'axile, parietal, free-central, etc.' (introduce terms after learners describe what they see; link to later fruit classification). For the CER say: 'A farmer in Kisumu tells you his single pawpaw tree flowers every season but never fruits.	Think-Pair-Share lets all learners in a large class verbalise observations of placentation before sharing, so quieter learners contribute. Claim-Evidence-Reasoning forces learners to link structural data (unisexual versus bisexual, ovary position) to the phenomenon, rather than only memorising terms.	Listen for pairs that name chambers and ovule attachment accurately. Use a CER rubric: claim answers the question; evidence cites at least 2 specific worksheet observations; reasoning links pollen from stamens to fertilisation of the ovary. Exit-ticket preview: learners write one sentence on how the maize plant differs from the Hibiscus in the position of male and female parts. Record the number of pairs (target 80%) whose reasoning includes the pollen-to-ovary link.

in the ARES library for this activity. Search ARES for readings Search terms: reproduction plants superior ovary	many flowers but no fruit?' Example claim: 'Flowers that are unisexual cannot make fruit alone.' Evidence: data from the worksheet (maize tassel only male, silks only female; Hibiscus bisexual). Reasoning: fruit develops from a fertilised ovary, so an ovary needs pollen from stamens; if the flowers or the tree lack one part, pollen must come from another flower or tree. Learners also link vegetative cuttings (Lesson 1) to the lack of any need for flowers or partners.	Using ONLY your evidence, what do you tell him?' Give 3 minutes of silent writing, then cold-call 3 learners to read a claim, and ask the class: 'Which evidence is strongest? What is missing?' Push reasoning: 'Why does having only male flowers on one tree stop fruit forming?' Correct the misconception that 'lots of flowers means lots of fruit' and introduce the terms bisexual, unisexual, monoecious (maize) and dioecious (some pawpaw trees). Close by stating: 'Hibiscus has both parts in one flower, so it could in principle pollinate itself. So why does a bisexual flower still sometimes not fruit alone? Hold that thought.'		
Driving Question Board (DQB) Creation  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: reproduction plants superior ovary  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: reproduction plants superior ovary	Learners return to the Driving Question Board (a chart or a section of the wall) with the main driving question: 'Why do some farm plants produce offspring without seeds while others require seeds and a partner tree to make fruit?' Each group writes 1-2 new questions on sticky notes or paper strips arising from today's dissection (for example 'If Hibiscus has both male and female parts, why does it need another flower?', 'Why does maize have separate male and female flowers?', 'Why is the passion flower so complicated?'). They place notes under categories: Structure of flowers, Pollination, Fruit and seed formation, Cuttings and asexual reproduction. They also mark earlier questions now partly answered, using a tick.	Say: 'What did you see today that you cannot explain yet? Write it as a question, not a fact.' WAIT TIME 10-15 seconds, then collect questions from at least 5 groups by cold-call. Highlight the question 'If flowers have both male and female parts, why don't all flowers pollinate themselves?' as the bridge to Lesson 3 (if learners do not raise it, pose it yourself and ask them to vote on whether it belongs on the board). Move 2 answered questions into an 'Answered so far' section with the evidence noted (for example 'Maize has separate male and female flowers: dissection, Lesson 2'). Make sure sticky-note authors come from different groups so every group contributes.	The Driving Question Board makes the class's growing understanding and remaining gaps visible, so learners see that today's evidence moves them toward the anchoring phenomenon and generates the next investigation.	Observable indicators: every group posts at least one question; at least half of the questions are testable or ask 'why/how' about structure and function and not just 'what is'; learners categorise questions accurately. Teacher notes which questions show confusion between pollination and fertilisation to address in Lesson 3.
Model Building	Learners revisit the class and individual models from Lesson 1	Say: 'Use a different colour. What new part of the story can your	Cumulative model revision lets learners integrate new structural	Oral probes: learners can state that pollen from anthers must

Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: reproduction plants superior ovary  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: reproduction plants superior ovary	(how plants reproduce: asexual cuttings versus sexual seeds). In groups they add a new 'Flower' section: a labelled diagram of a bisexual flower with stamens and carpel, a small sketch showing maize's separate tassel and silk, and an arrow system showing that fruit needs pollen to reach the ovary. They use a different colour pen for today's revisions, and add a 'Still unsure' box with the question: 'If flowers have both parts, why don't they all pollinate themselves?' Each group explains the revised model to a neighbouring group. Individually they complete an exit ticket.	model now explain about the lone avocado tree?' WAIT TIME 10-15 seconds before groups begin. Circulate for oral probes, asking at least 6 groups: 'What does your model show about where pollen is made and where it must go? What would happen to fruit if a tree only had male flowers? What is still missing from the model?' Cold-call 2 groups to present a revision. Exit ticket (written on a half sheet, 3 minutes): 1) Name one difference between a superior and an inferior ovary. 2) Give one example of a unisexual flower arrangement from a Kenyan crop. 3) Write one question about why a flower might not pollinate itself. Collect tickets at the door and count the blades one last time before dismissal. Remind learners to wash their hands. Announce: 'Next lesson we investigate why flowers with both parts still avoid fertilising themselves.'	evidence into their explanation of the phenomenon, and the 'Still unsure' box turns a gap into purpose for the next lesson.	reach the stigma and that unisexual flowers or separate trees change where pollen comes from. Model check: revised models show labelled stamen and carpel, include maize tassel and silk, and have a visible new question. Exit ticket success criteria: at least 70% correctly define superior versus inferior ovary and give a correct crop example (maize, pawpaw, Hibiscus). Use results to plan re-teaching for the next lesson's starter.
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D. TEACHER REFLECTION

1) Did every group get a turn using the blade and hand lens, and did safety protocols hold with no injuries and a correct blade count? 2) Which learners still think every flower is bisexual or that more flowers always mean more fruit? 3) Were specimens fresh and available for all four species, and what substitutes (tomato, bean, pawpaw, sweet potato flowers, or cached photos) worked well? 4) Which groups struggled to decide ovary position, and would pre-sliced demonstration flowers help? 5) How well did the DQB questions lead toward self-pollination, and which learners raised it unprompted? 6) What would you change in timing for a class of 45-50 learners?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We dissected Hibiscus, Bougainvillea, maize inflorescence and passion flower. We saw that Hibiscus is bisexual, with stamens and carpel in one flower; maize has separate male (tassel) and female (silk/cob) flowers on the same plant; Bougainvillea has small flowers surrounded by showy bracts; and flowers differ in symmetry, ovary position and arrangement in clusters.
What did I learn?	Flowers can be classified by ovary position (superior or inferior), symmetry (actinomorphic or zygomorphic), inflorescence (single or clustered) and by whether they are bisexual or unisexual. The ovary holds the ovules, and its structure shows how the seeds are attached (placentation).
How does this explain the	A lone pawpaw or avocado tree may flower heavily but not fruit because fruit needs pollen from stamens to reach the ovary of a

phenomenon?	carpel. If the tree's flowers are unisexual or lack the matching partner parts, pollen must come from another flower or tree. Cassava and sweet potato cuttings skip flowers altogether. New question: if flowers have both male and female parts, why don't all flowers pollinate themselves?
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LESSON 3 (80 minutes (double lesson)): Why Can a Lone Pawpaw Tree Flower Heavily but Never Set Fruit? Adaptations That Prevent Self-Pollination

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students gather evidence that flowering plants have structural, timing and chemical adaptations that hinder self-pollination and self-fertilisation. This explains why a lone pawpaw or avocado tree can be covered in flowers yet bear no fruit, while cassava and sweet potato cuttings need no partner at all.
Knowledge	By the end of the lesson the learner should be able to: (1) define self-pollination and cross-pollination; (2) describe protandry, protogyny, heterostyly, dioecy and self-incompatibility, with a Kenyan example of each where possible (e.g. protandry in maize tassels/silks and Hibiscus, dioecy in pawpaw, heterostyly in Primula-type flowers, self-incompatibility in passion fruit); (3) explain how each adaptation promotes cross-pollination; (4) link these adaptations to the need for a 'partner tree' in fruiting.
Skills	NGSS SEP: Developing and Using Models (building and using wire/bead models and charts to trace pollen movement); Planning and Carrying Out Investigations (simulating structural hindrances); Engaging in Argument from Evidence (CER). Core competencies: Communication and Collaboration (cooperating with others while using models and charts); Critical Thinking and Problem Solving (analysing causes and effects of failed fruiting in a lone tree).
Attitudes	Values: Love - the learner prioritises others' interests while using models and charts to illustrate mechanisms and features that hinder self-pollination and self-fertilisation (e.g. ensuring every group member handles the model, speaks and is heard). PCI: Environment and Technology - appreciation of how knowledge of pollination helps farmers manage orchards and the environment (e.g. planting male and female pawpaw together).
Key Inquiry Question	KICD Key Inquiry Questions: 1. What is the significance of reproduction in plants? 2. How does reproduction occur in plants? Lesson focus: Why does a lone tree with many flowers fail to produce fruit, and what features of the flower prevent it from pollinating itself?
Purpose in Storyline	Lessons 1-2 established that asexual reproduction (cuttings of cassava, sugarcane, sweet potato) needs no partner, and that flowers have male and female parts and can be classified. Lesson 3 explains the other half of the puzzle: sexual reproduction in many crops actively avoids self-pollination, so a lone flowering tree may stay barren. It leads to the next question: What actually happens inside the flower after pollen successfully lands on a receptive stigma?
Safety Notes	Wire models: wire ends must be bent into loops or capped with beads so no sharp ends are exposed; the teacher pre-cuts and pre-prepares the wire. Hand lenses: never look at the sun through them. Plant material: check for allergies to pollen (Hibiscus, maize tassels), and for asthma; students with hay fever may observe from a distance or wear a cloth mask. Wash hands after handling plant material. If scalpels or razor blades are used for any optional flower sectioning, only the teacher or a designated group member may handle them, in groups of 4-5, blade cutting away from the body, on a tray, and blades are counted out and counted back in (full blade protocols are detailed in Lessons 2 and 5). This lesson is designed to run primarily with models, charts and whole flowers that are pulled apart by hand, so that no blades are needed. Do not taste any plant material in the lab.

B. LESSON OVERVIEW

Students begin with the puzzle of a lone pawpaw tree full of flowers that never sets fruit and predict why. They then rotate through model-based stations in groups of 4-5 (45-50 students per stream), using flower structural charts and wire/bead models of styles and stamens to trace pollen movement and simulate structural hindrances: protandry (maize tassels shed pollen before silks are receptive), protogyny, heterostyly (pin and thrum flowers), dioecy (male and female pawpaw trees), and self-

incompatibility (passion fruit pollen rejected by its own stigma). Using Predict-Observe-Explain, Think-Pair-Share and Claim-Evidence-Reasoning, students build explanations, update the Driving Question Board (DQB) and revise their class model. They leave with a new question about what happens inside the flower after compatible pollen lands.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: reproduction plants protandry protogyny  EXERCISE: Apply: plant reproductive success Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Students look at a photo or drawing of a lone pawpaw/avocado tree in a compound in Kisumu covered in flowers but with no fruit, next to a fruiting tree in a shamba with neighbouring trees. Individually, they write a prediction in their notebooks: 'Why does the lone tree not set fruit even though pollen and ovules may both be present?' and sketch what they think stops a flower from fertilising itself. They recall Lesson 2 flower parts (anther, filament, stigma, style, ovary). They then share predictions in pairs and the class lists ideas on the board (e.g. 'pollen is not mature', 'flowers are only male or only female', 'the stigma rejects the pollen', 'parts are too far apart').	Display the picture. Say: 'This pawpaw has hundreds of flowers but not one fruit. Yet a cassava cutting grows with no flower at all. What is going on?' Give 1 minute of silent WAIT TIME (at least 10-15 seconds before any hands go up) for individual writing. Say: 'Write your prediction and one reason. There are no wrong predictions today, only ideas to test.' Cold-call 4 students from different parts of the room to share, then ask 2 follow-ups: 'What would we need to see to know your idea is right?' and 'Who has a different idea?' Record all predictions on a 'Our Predictions' chart without judging. Circulate to check that quieter students and those at the back of the large class are writing. Link back: 'Remember our phenomenon: a piece of stem can become a whole plant, but a flowering tree may need a partner. Today we find out why.'	Predict-Observe-Explain (POE) - Predict step. Writing predictions before seeing evidence surfaces prior ideas and misconceptions (for example 'a flower always pollinates itself') and gives students a reason to care about the evidence they gather. They can compare their later observations against their own prediction, which builds ownership of the explanation.	Observable indicators: Can each student write a prediction with a reason? Do at least 80% of students name a plausible cause related to flower parts, timing or separate sexes? Teacher tallies the types of predictions (timing, structure, separate sexes, chemical rejection, 'no pollinators') on a clipboard and notes common misconceptions (e.g. 'the tree is sick'). Thumbs check: 'Raise a finger for how confident you are in your prediction: 1 to 3.'
Observe Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: reproduction plants protandry protogyny	In groups of 4-5 students rotate through four stations, about 6 minutes each, with roles (Model Handler, Recorder, Reporter, Materials Monitor; roles rotate so everyone handles the model). Station 1 - Timing (protandry/protogyny): Students use a labelled chart and fresh or pictured maize (tassels shed pollen before silks emerge on the	Before the lesson, set up four stations on desks around the room, with 2 sets of each station to manage 45-50 students (around 10-11 groups): charts, wire/bead models, hand lenses (shared, 1 per group of 4-5) and fresh Hibiscus and maize if available. Brief for 3 minutes: 'First predict, then observe. At each station, write what you PREDICT will	Predict-Observe-Explain (POE) - Observe step, with Developing and Using Models. Handling physical models lets students 'see' invisible processes: pollen travelling, timing mismatch, and chemical rejection. Predicting before moving each bead creates cognitive conflict when the model contradicts the idea that 'a flower pollinates itself', and the observation table forces	Teacher checks each group's observation table and labelled diagrams. Observable indicators: (a) the group correctly labels anther, filament, stigma, style and ovary on at least one model; (b) the table lists all five adaptations with a correct Kenyan example for at least four; (c) the group can show with the model why pollen fails to reach the stigma in at least

 EXERCISE: Understand: plant reproductive success Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	cob) and Hibiscus flowers (anthers shed pollen before the stigma lobes become receptive) to trace pollen movement and complete a timeline showing 'pollen ready' versus 'stigma ready'. They record why a single flower cannot pollinate itself if the times do not overlap, and sketch protogyny as the reverse. Station 2 - Structure (heterostyly and position of parts): Using wire/bean models of styles and stamens, students build a 'pin' flower (long style, short stamens) and a 'thrum' flower (short style, long stamens), then move a bead 'pollen grain' from anther to stigma to see that pollen from the same flower cannot reach the stigma but pollen carried by an insect from a different-height anther can. They compare with a Hibiscus or tomato flower, where the stigma projects beyond the anthers. Station 3 - Separate sexes (dioecy): Using pictures/models of male pawpaw flowers (long stalks, many stamens) and female flowers (large ovary, no functional stamens), students sort flowers from 'tree A' and 'tree B', and trace on a chart where pollen must travel for a fruit to form. They compare with maize (monoecious: separate male tassels and female cobs on the same plant) to see that separate flowers reduce but do not eliminate self-pollination. Station 4 - Chemistry (self-incompatibility): Using a coloured-bead simulation with passion fruit (pollen beads and stigma cards marked with matching 'S-type' colour codes), students try to 'germinate' pollen on a stigma card: if the code matches the stigma's own code, the pollen is rejected; if different, it is accepted.	happen before you move the bead or the card.' Reinforce safety: 'Bent wire ends only. Hand lenses never face the sun. Hands washed afterwards.' Circulate with a checklist. Ask probing questions at each station: 'Where is the pollen when the stigma is ready?'; 'What would happen if the bee only visited this one flower?'; 'How is this different from a cassava cutting?' After posing a question, use 10-15 seconds WAIT TIME before taking answers. Cold-call 2 students per station visit, e.g. 'Wanjiru, explain what the bead movement shows.' Call time every 6 minutes with a bell. If a group finishes early, challenge them: 'Design a flower that could never pollinate itself using two adaptations.' For students who struggle, supply a labelled flower chart and sentence starters: 'The pollen cannot reach the stigma because...'	students to connect a structure to its function.	two adaptations; (d) all members take a turn handling the model (Love value - prioritising others' interests). Oral probes at each station: 'Show me where the pollen would need to travel.' Record ticks/crosses on a class list.
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	They record how many of 10 trials succeed from the same flower versus a different flower. Each group completes a one-page observation table: adaptation, plant example, how it prevents self-pollination, what a farmer must do to get fruit.			
Explain Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: reproduction plants protandry protogyny  EXERCISE: Apply: plant reproductive success Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Part A - Think-Pair-Share (8 minutes): Students individually think about the question 'Which adaptation best explains the lone pawpaw that flowers but never fruits, and which explains why cassava cuttings need no partner?' They pair up, compare and then share with the class. Part B - Claim-Evidence-Reasoning (12 minutes): Each student writes a CER paragraph in their notebook: Claim - 'The lone pawpaw tree fails to set fruit because...' (e.g. it is a male or female tree only - dioecy - so the flowers cannot pollinate themselves); Evidence - cite data from their station observation tables (e.g. 'In the Station 3 sorting, tree A had only male flowers with stamens, so no ovary could form fruit'); Reasoning - connect the evidence to the principle that these adaptations promote cross-pollination and genetic variation, which is useful for the species (e.g. varied maize and bean crops resist disease) but a problem for a farmer with one tree. They also compare with sexual versus asexual reproduction: cassava cuttings give identical clones with no need for partners, whereas fruiting trees such as passion fruit often need cross-pollination. Students peer review a partner's CER using a 3-point rubric (claim clear, evidence from stations, reasoning links	Pose the question for Think-Pair-Share: 'Which adaptation explains the lone pawpaw?' Give 1 minute silent think time (WAIT TIME 10-15 seconds before taking any response). Pair for 2 minutes, then cold-call 5 pairs: 'Kamau and Achieng, what did you decide, and what evidence supports it?' Say: 'Use the words from our stations: stigma, anther, protandry, dioecy.' Address misconceptions explicitly, e.g. 'Does cross-pollination mean the flower is weak?' or 'Is self-pollination always bad?' Clarify that some crops (beans, tomatoes) are largely self-pollinating, which is why bean farmers can save seed, while others avoid it. Model one CER sentence on the board, then let students write. Display the rubric and walk around reading 8-10 CERs. Say: 'Swap notebooks. Give one glow and one grow using the rubric.' Then cold-call 3 students to read their partner's best sentence. Remind that sharing the rubric and feedback fairly is a Love value in action.	Think-Pair-Share (TPS) followed by Claim-Evidence-Reasoning (CER). TPS gives every student in a large class thinking time and a low-risk rehearsal of ideas before speaking, ensuring the quieter students contribute. CER then structures argument from evidence: students must tie their claim to the data from their models, which turns observations into a causal explanation that answers part of the driving question.	Peer review rubric (3 points: claim, evidence, reasoning) scored by partners and spot-checked by the teacher. Observable indicators: claim states a specific adaptation; evidence cites at least one observation from the stations; reasoning explains how the adaptation promotes cross-pollination. Success criterion: 70% of students score 2 or 3. Teacher notes pairs that confuse dioecy with monoecy or that think self-incompatibility is about the soil, for reteaching next lesson.

	structure to function) and offer one 'glow' and one 'grow'.			
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: reproduction plants protandry protogyny  EXERCISE: Apply: plant reproductive success Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Students return to the class Driving Question Board. Working in their groups, they write on sticky notes (a) one thing they have now figured out about why some plants need a partner tree (e.g. 'Pawpaw trees are male or female, so the lone tree has no partner') and place it under 'What we figured out'; and (b) one new question raised by today's work (e.g. 'Once compatible pollen lands on the stigma, how does it reach the ovule?', 'What happens inside the flower after pollen lands?', 'Why do passion fruit farmers in Kericho hand-pollinate?') and place it under 'New questions'. They also move or retire earlier questions that have been answered, such as 'Why does a lone tree have flowers but no fruit?'.	Say: 'Look back at the question we started with. What can we now answer? What new question has appeared?' Give 10-15 seconds WAIT TIME while students think and write. Ask each group to send one member to the board (so about 10 students move, not 50). Cold-call 3 groups to read one sticky note aloud and ask: 'Which part of today's evidence led to that question?' Group similar questions together and highlight the question that leads to the next lesson: 'What actually happens inside the flower after pollen successfully lands on a receptive stigma?' Say: 'That is where we go next.' Photograph the DQB so progress can be kept if the board is cleaned. If there is no wall space, use a chart paper or a section of the blackboard.	Driving Question Board (DQB). The DQB makes the class's thinking visible and shows how each lesson's evidence answers part of the driving question while generating new questions. Students see they have a role in shaping the unit's direction, which sustains curiosity and motivation and builds coherence between lessons.	Exit ticket (completed on a slip, 3 minutes, can be on DQB-sticky-note paper): 1) Name two adaptations that prevent self-pollination and give a Kenyan plant example for each. 2) Why might a lone pawpaw tree not bear fruit? 3) One question you still have. Observable indicators: sticky notes cite specific evidence; at least one new question per group relates to events after pollination; questions are testable rather than yes/no. Teacher sorts exit tickets into 'got it / nearly / not yet' piles for the next lesson's starter.
Model Building Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: reproduction plants protandry protogyny  EXERCISE: Apply: plant reproductive success Source: Khan Academy (English - US curriculum)  Search ARES	Students return to their initial class model (begun in Lessons 1-2) that explains why some plants reproduce without seeds and others need seeds and a partner. In groups they revise their model on chart paper: they add a new section for 'Barriers to self-pollination' with labelled drawings of the five adaptations (protandry/protogyny timing graph, heterostyly pin/thrum diagram, dioecy male/female pawpaw trees, self-incompatibility passion fruit rejection arrow) and add arrows showing pollen being carried by wind (maize) or insects (Hibiscus, passion fruit - carpenter bees) to a different plant. They mark with a question mark the 'black box' of what happens after pollen lands on	Say: 'Take out your model from last lesson. What needs to be added or changed based on today's evidence? Use a different coloured pen for new ideas so we can see how your thinking has grown.' Circulate and ask oral probes: 'Show me on your model why the lone pawpaw has no fruit.' 'Where does your model explain the cassava cutting?' 'What is still missing from your model?' Use 10-15 seconds WAIT TIME after each question. Cold-call at least 5 groups to explain one revision, with each student in the group contributing a sentence. During the gallery walk, set a 4-minute timer: 'Two minutes at each model, one helpful suggestion per group.' Conclude: 'Our model explains	Cumulative Model Building (Developing and Using Models). Revising a shared model across lessons lets students integrate new evidence into a coherent explanatory framework and see gaps in their understanding. Using a different pen colour makes conceptual change visible, and the question mark showing what is unexplained links directly to the next lesson.	Labeled diagram check and oral probes during model building. Observable indicators: (a) model correctly labels anther, stigma, style and ovary; (b) includes at least three barriers to self-pollination with correct examples; (c) shows pollen movement with arrows between different plants; (d) explains the lone pawpaw case in one sentence; (e) identifies the unexplained step after pollen lands. Teacher uses a quick checklist (yes / partially / not yet) per group and plans targeted support for groups scoring 'not yet' on (a) or (c).

for similar readings	a compatible stigma, which will be investigated in Lesson 4. They also add a note comparing farmers' choices: cuttings (cassava, sweet potato) bypass these barriers; seed crops depend on them. Groups do a short gallery walk of 2 other models, leaving one sticky-note suggestion on each.	why a partner tree is needed, but it cannot yet explain what happens after the pollen lands. That is our next job.' Collect exit tickets at the door and review diagrams.		
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D. TEACHER REFLECTION
After the lesson, reflect: (1) Did every student handle a model, or did a few dominate with 4-5 per group? Adjust roles or add more model sets next time. (2) Which adaptation did students find hardest to understand (likely self-incompatibility, since it is invisible), and would a larger bead simulation help? (3) Were common misconceptions addressed, such as confusing dioecy with monoecy, thinking self-pollination is always harmful, or assuming pollination equals fertilisation? (4) Did stations run on time with large classes, and were safety rules for wire ends and hand lenses followed? (5) What do the exit tickets show about students' ability to link the lone pawpaw case to the specific adaptations? (6) Which Kenyan examples resonated most (maize, Hibiscus, pawpaw, passion fruit), and are there local examples from the students' home areas to add? (7) Is the key question for Lesson 4 - what happens inside the flower after pollen lands - clear and motivating to students? Note the blade-safety protocols must be fully rehearsed before Lessons 2 and 5 where dissection takes place.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We used wire/bead models and flower charts to trace pollen movement in maize, Hibiscus, pawpaw and passion fruit. We saw that pollen and stigma can be ready at different times (protandry/protogyny), that styles and stamens can be at different heights (heterostyly), that some trees have only male or only female flowers (dioecy), and that some stigmas reject their own pollen (self-incompatibility).
What did I learn?	Many flowering plants have timing, structural, separate-sex and chemical adaptations that prevent self-pollination and promote cross-pollination, so a lone tree like a pawpaw or passion fruit may flower but not set fruit without a compatible partner. Plants propagated by cuttings (cassava, sweet potato, sugarcane) bypass these barriers because no pollination is needed.
How does this explain the phenomenon?	A lone pawpaw tree covered in flowers can stay fruitless because it is male or female only (dioecy), or because its flowers cannot pollinate themselves, so a partner tree is needed to supply compatible pollen. A cassava or sweet potato cutting grows into a new plant without flowers or a partner because it reproduces asexually. Next question: What actually happens inside the flower after pollen successfully lands on a receptive stigma?

LESSON 4 (80 minutes (double lesson)): Following the Pollen Tube: How Double Fertilisation Explains Why a Lone Pawpaw Flowers but Sets No Fruit

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use cached animations of pollen tube growth to trace the pathway of male nuclei from pollen grain to embryo sac, and explain double fertilisation (one sperm nucleus fuses with the egg cell to form the zygote, the other fuses with the polar nuclei to form the endosperm). They then use this to explain why flowering plants such as maize, beans and passion fruit need successful fertilisation to make seeds and fruits, unlike cassava or sweet potato cuttings.
Knowledge	By the end of the lesson the learner should be able to: (1) describe the process of double fertilisation in flowering plants (SLO d); (2) state the fate of the zygote (embryo) and of the fused polar nuclei (endosperm); (3) link fertilisation to the need for pollination and a compatible partner tree (pawpaw, avocado); (4) contrast this with asexual propagation where no fusion of gametes occurs.
Skills	NGSS Practices: Developing and Using Models (diagramming the pollen tube pathway), Analyzing and Interpreting Data (extracting evidence from animation frames), Constructing Explanations (CER on the lone-tree puzzle), Engaging in Argument from Evidence. Digital literacy: using cached animations safely and critically.
Attitudes	Core competencies: Communication and Collaboration (cooperating in groups of 4-5 while tracing and checking pathway diagrams); Critical Thinking and Problem Solving (analysing causes and effects of failed fertilisation). Values: Social Justice (sharing devices, hand lenses and the single projector equitably); Love (prioritising group-mates' understanding when peer reviewing). PCIs: Environment and Technology (appreciating animations as a tool for seeing hidden processes); Online Safety (using only teacher-cached, KICD-approved resources and discussing why unknown sites are avoided).
Key Inquiry Question	1. What is the significance of reproduction in plants? 2. How does reproduction occur in plants?
Purpose in Storyline	Lessons 1-3 established how cassava, sugarcane and sweet potato regrow from cuttings without seeds (asexual), and how flower structure and features hindering self-pollination explain why a lone pawpaw or avocado may fail to set fruit. Lesson 4 supplies the missing mechanism inside the flower: what actually happens after pollen lands, and why fusion of nuclei is the trigger for seed and fruit formation. It ends with a new question that launches Lesson 5: How does double fertilisation transform the ovary and ovules into fruits and seeds?
Safety Notes	This lesson is mainly digital and diagramming, but a short optional hand-lens look at a hibiscus or maize flower may be used. Scalpels and razor blades are NOT used in this lesson (dissection safety protocols are covered explicitly in Lessons 2 and 5). Teacher keeps all blades locked away. Hand lenses: share in groups of 4-5, never look at the sun through a lens, carry with both hands. Devices: use only the teacher-cached animation files; no learner browsing online. Keep cables off walkways in a crowded classroom of 45-50 learners. Wash hands after handling flowers (some, e.g. pollen from hibiscus, can cause sneezing; learners with allergies observe from a distance or use a diagram).

B. LESSON OVERVIEW

Learners return to the anchoring puzzle: a lone pawpaw or avocado tree covered in flowers but with no fruit, versus a cassava stem that becomes a new plant with no flower at all. They predict what must happen inside a flower for a fruit to form, then watch cached animations of pollen germination, pollen tube growth through the style, entry through the micropyle and the two fusions of double fertilisation. In groups of 4-5 they trace and label the pathway on a worksheet, use Think-Pair-Share to interpret what each fusion produces (zygote and triploid endosperm, as in the maize grain used for ugali), and write a Claim-Evidence-Reasoning explanation for why a lone tree fails to fruit. The class updates the Driving Question Board and revises its emerging model, ending with a new question about how the ovary and ovules

become fruit and seed.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Fertilization terminology: gametes, zygotes, haploid, diploid Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Apply: plant reproductive success Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners look at a projected photo/sketch (or teacher-drawn board sketch) of a flowering pawpaw tree with no fruit beside a cassava cutting sprouting. Individually in workbooks they write and sketch a prediction (3 min): 'What has to happen inside a flower, after pollen lands, for a seed and fruit to form?' They draw one arrow-diagram prediction of the route of the male cell. They then share with a partner (3 min) and a few predictions are collected. Learners recall earlier evidence from Lessons 1-3 (asexual propagation of cassava, sugarcane, sweet potato; features hindering self-pollination such as separate male and female pawpaw trees).	Say: 'Think of the lone pawpaw in a compound in Kisumu, full of flowers but no fruit. Pollen may even land on it. What is missing?' Give 3 minutes silent writing, then WAIT TIME of 10-15 seconds before taking any answer. Cold-call 5 learners from different parts of the room (use the class register, not volunteers) and ask: 'What did you draw for the route of the male cell, and why?' Record 3 distinct predictions on the board without judging (e.g. 'pollen goes straight to the ovary', 'pollen makes a tube', 'pollen mixes with nectar'). Ask: 'Which of these could we test with evidence today?' Collect the prediction sketches for later comparison. Circulate to check that learners with fewer devices or less confidence are included in pairs.	Predict-Observe-Explain (POE): committing to a written, drawn prediction first exposes prior ideas and misconceptions (for example that pollen lands directly on the egg), creating a need to know that makes the animation purposeful. Learners later compare their own prediction with the evidence, building ownership of the explanation.	Observable indicators: (a) each learner has a dated written prediction plus arrow sketch in the workbook (teacher scans 15 workbooks while circulating); (b) at least 3 different predictions recorded on the board; (c) tally misconceptions such as 'pollen falls directly onto the ovule' or 'fertilisation = pollination' to address in the Explain phase. Look for learners who connect prediction to the lone-tree phenomenon.
Observe Phase  VIDEO: Fertilization terminology: gametes, zygotes, haploid, diploid Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Understand: plant reproductive success	Learners watch the cached KICD-approved animation of pollen germination and pollen tube growth and double fertilisation on the projector/shared devices (3 viewings: first full run for the story, second paused at key frames, third with groups tracing). Using the pathway diagram worksheet, groups of 4-5 trace and label the route: pollen grain lands on stigma, pollen grain germinates, tube nucleus and generative nucleus, tube grows down the style, enters the ovary, ovule and embryo sac through the micropyle,	Before playing: 'While you watch, trace the male nucleus with your finger on the worksheet. Where does it start, where does it end, and what does it meet?' Remind about Online Safety: 'We only use the cached KICD-approved file. Why do we avoid opening unknown links on the school device?' Play the full animation once without stopping. Then replay and pause at 4 frames: stigma landing, tube in style, micropyle entry, the two fusions. At each pause ask: 'What do you see happening to the nuclei right now?'	Evidence gathering with Developing and Using Models: the animation lets learners see a process that is invisible in whole flowers. Pausing at key frames and tracing the pathway links each structure to its function and turns a passive video into data that tests the predictions from the Predict phase.	Labelled pathway diagram check: observable indicators on the worksheet are (1) correct sequence stigma, style, ovary, ovule, micropyle, embryo sac; (2) label of both sperm nuclei; (3) correct products: zygote and endosperm; (4) arrows drawn in the correct direction. Teacher marks a quick tick/cross on 1 diagram per group during circulation and notes groups needing re-teaching of the micropyle or polar nuclei. Oral probe: 'Show me where the second sperm nucleus goes.'

Source: Khan Academy (English - US curriculum) Search ARES for similar readings	two sperm nuclei released. They record in an evidence table: structure, what happens there, and what the animation showed. They note the two fusions: sperm nucleus 1 + egg cell = zygote (diploid); sperm nucleus 2 + two polar nuclei = triploid endosperm nucleus. Roles in each group: Device/Projector Monitor (equitable turns), Tracer, Labeller, Reporter. Optional 5-minute hand-lens look at a hibiscus or maize flower to locate stigma, style and ovary (no blades).	and allow WAIT TIME of 10-15 seconds before cold-calling 2 learners per pause (8 total). Circulate, using a checklist to scan each group's worksheet; prompt: 'Which structure does the tube pass through before it meets the egg?' and 'How many sperm nuclei are released, and what does each one join?' Stress that the pawpaw puzzle needs this: if pollen from a compatible partner tree never arrives, there is no tube and no fusion. For large class management: place 2 groups per cached device or rotate through the projector; keep the hand-lens station to one group at a time on rotation. Ask learners who finish early to add arrows showing the haploid/diploid/triploid number of each product.		
Explain Phase  VIDEO: Fertilization terminology: gametes, zygotes, haploid, diploid Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Apply: plant reproductive success Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners compare the animation evidence with their predictions and revise their sketches in a different colour. Think-Pair-Share: 'Why is it called double fertilisation, and why might it be an advantage for a plant like maize or beans to fuse the second nucleus with polar nuclei?' Pairs discuss the endosperm as food store (the white starchy part of the maize grain that becomes ugali flour; the cotyledon-store in beans) and the triploid product. Then learners write an individual Claim-Evidence-Reasoning (CER) paragraph: 'Claim: A lone pawpaw/avocado tree flowers heavily but sets no fruit because ...' using the animation frames as evidence and fertilisation as the reasoning. They contrast with cassava, which regrows from a cutting with no gamete fusion (asexual), and state	Say: 'Look at your first prediction. Which parts did the animation support, and which parts need changing?' Give 2 minutes then WAIT TIME 10-15 seconds before cold-calling 4 learners to share one change. Launch Think-Pair-Share: 1 minute think, 3 minutes pair talk, then cold-call 5 pairs, asking different pairs to add rather than repeat. Press with: 'What evidence from the animation shows that the second fusion is separate from the first?' and 'What would happen to the seed if the endosperm did not form?' Correct misconceptions: pollination is not fertilisation; pollination is the transfer, fertilisation is the fusion. Provide the CER sentence frames on the board: 'I claim ... The evidence is ... This shows ... because ...'. Circulate and read 10 CER statements; select 2 (with permission) to read aloud and ask	Think-Pair-Share lets all 45-50 learners rehearse ideas aloud in a large class, and Claim-Evidence-Reasoning forces them to connect observed animation evidence to the scientific principle of gamete fusion, so that the explanation of the lone-tree puzzle is grounded in evidence rather than recall.	Observable indicators: CER statements that (1) claim failure of pollination/fertilisation due to absence of compatible pollen; (2) cite evidence such as pollen tube growth and sperm nuclei fusion; (3) reason that no fusion means no zygote or endosperm, so no seed and no fruit; (4) correctly contrast with asexual cuttings. Use a 3-point check (claim, evidence, reasoning) while circulating. Listen in pairs for correct use of 'triploid', 'zygote' and 'endosperm'.

	one advantage and one disadvantage of each method for a Kenyan farmer (e.g. cassava cuttings are fast and identical to the parent, while seeds from maize give variation and can be stored).	the class to identify the claim, evidence and reasoning. Tie to the phenomenon: 'So why does a lone pawpaw with a thousand flowers still give nothing, while a cassava stem needs no flower at all?'		
Driving Question Board (DQB) Creation  VIDEO: Fertilization terminology: gametes, zygotes, haploid, diploid Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Apply: plant reproductive success Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups add sticky notes or chart-paper entries to the Driving Question Board: (a) What we now understand (e.g. 'Pollen tubes carry two sperm nuclei to the embryo sac; one forms the zygote, one forms the endosperm'), (b) Questions answered from earlier lessons, and (c) New questions (e.g. 'What happens to the ovary and ovules after fertilisation?', 'Why does a passion fruit have so many seeds?', 'Does every flower with pollen make a fruit?'). Learners complete an exit ticket individually: 'One new thing I learned; one question I still have.'	Say: 'Go back to the driving question: why do some plants need seeds and a partner tree? What can we now add to the board?' Give each group 3 minutes to write 2 sticky notes, then WAIT TIME 10-15 seconds as groups review the board. Cold-call 3 groups to read a new question aloud and ask: 'Which lesson do you think will help answer this?' Cluster the questions and circle the one that leads to Lesson 5: 'How does double fertilisation transform the ovary and ovules into fruits and seeds?' Hand out exit tickets and collect them at the door. Remind learners to bring a tomato, passion fruit or bean pod (or sketch from home) for Lesson 5, and to share pods equitably within groups.	The Driving Question Board makes learning visible and public: each lesson's evidence is linked to the anchoring phenomenon and to a new question, so learners see the storyline progress and own the purpose of the next lesson.	Exit ticket check: indicators are (1) correct statement of double fertilisation (two fusions, two products); (2) a genuine new question connected to fruit or seed formation. Sort tickets into three piles (secure, partial, needs reteach) and use them to plan the next lesson's starter. DQB check: at least one new question per group that relates to the phenomenon.
Model Building Phase  VIDEO: Fertilization terminology: gametes, zygotes, haploid, diploid Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Apply: plant reproductive	Learners revise the class's cumulative model of 'How do plants reproduce with and without seeds?' using a large poster (or workbook page with two columns). The Asexual branch (cassava, sugarcane, sweet potato cuttings: no gametes, offspring identical to parent) and the Sexual branch are now extended to include inside-the-flower steps: pollination, pollen germination, pollen tube growth through style, entry through the micropyle, double fertilisation, zygote, endosperm. They add a box 'Why a lone pawpaw/avocado fails to set fruit: no compatible	Say: 'Let's add to our model. What new part must we draw between pollination and seed?' Give WAIT TIME of 10-15 seconds, then cold-call 4 learners to add one step each to the class poster. During pair work, ask oral probes: 'Why is the endosperm triploid?', 'If the pollen tube failed to reach the micropyle, what would you see in the fruit?' and 'How does your model explain why maize needs seeds but cassava does not?' Distribute the peer-review rubric (2 criteria per pair: 'accurate pathway' and 'links to phenomenon') and have pairs swap workbooks for 3	Cumulative model revision lets learners integrate asexual and sexual evidence into one coherent explanatory model and see how each lesson's evidence refines it; peer review builds communication, collaboration and love (prioritising a partner's understanding).	Oral probes during model building with observable indicators: learner can (1) name the two products of double fertilisation; (2) explain the failure of fruit set on a lone tree using the pathway; (3) differentiate pollination from fertilisation. Peer review rubric (accurate pathway, correct products, link to phenomenon): record counts of learners scoring 3/3, 2/3, 1/3 to plan Lesson 5 starter.

SUCCESS Source: Khan Academy (English - US curriculum) Search ARES for similar readings	pollen, so no pollen tube, so no fertilisation'. They also add a 'Still to explain' arrow pointing to fruit and seed formation (Lesson 5). Pairs peer-review each other's model using a quick rubric: accuracy of labels, correct pathway, correct products, link to phenomenon.	minutes, giving one star and one wish. Circulate to check 8-10 models and identify learners ready for extension (e.g. why self-incompatibility helps variation). Close by reading the next question aloud: 'How does double fertilisation transform the ovary and ovules into fruits and seeds?' Collect worksheets for marking.		
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D. TEACHER REFLECTION

After the lesson, consider: (1) How many learners could correctly state both products of double fertilisation without prompting? (2) Which misconceptions persisted (pollination = fertilisation, pollen reaching egg directly, endosperm as diploid)? (3) Did the cached animation run reliably and did the sharing of devices across 45-50 learners remain equitable? What would you change (more pauses, printed frame cards)? (4) Did learners connect fertilisation back to the lone pawpaw and the cassava cutting, or did the lesson feel like abstract diagramming? (5) Which exit-ticket questions can seed Lesson 5 on fruits, seeds and placentation? (6) Remind yourself to prepare Lesson 5 safety briefing on handling blades/razors and to ask learners to bring tomatoes, passion fruits and bean pods.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We watched cached animations showing a pollen grain land on a stigma, germinate and grow a pollen tube down the style to the micropyle of the ovule, releasing two sperm nuclei into the embryo sac. We traced and labelled this pathway on our worksheets.
What did I learn?	Double fertilisation: one sperm nucleus fuses with the egg cell to form the zygote (the future embryo), and the other fuses with the two polar nuclei to form the triploid endosperm (food store, as in maize grain for ugali). Pollination is the transfer of pollen; fertilisation is the fusion of nuclei.
How does this explain the phenomenon?	A lone pawpaw or avocado tree can be full of flowers but set no fruit because without compatible pollen from a partner tree no pollen tube grows, no fusion occurs, and no zygote or endosperm forms, so no seed or fruit develops. Cassava, sugarcane and sweet potato cuttings need no fusion of gametes, so they grow into new plants without flowers or seeds. New question: How does double fertilisation transform the ovary and ovules into fruits and seeds?

LESSON 5 (80 minutes (double lesson)): Inside the Fruit: Placentation as Evidence of Post-Fertilisation Development

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners gather physical evidence that fruits and seeds develop from the ovary and ovules after fertilisation. They section local fruits and classify placentation, which helps explain why a lone flowering tree can bear no fruit while a vegetative cutting needs no flower at all.
Knowledge	By the end of the lesson the learner should be able to: (1) state that after fertilisation the ovule becomes the seed and the ovary becomes the fruit; (2) define placentation and distinguish marginal, axile, parietal and free-central types; (3) classify local fruits by placentation using evidence from sections.
Skills	SEP 3 Planning and Carrying Out Investigations: safely section fruits and record observations. SEP 2 Developing and Using Models: draw and label sectioned fruits and revise the class model. SEP 6 Constructing Explanations: use Claim-Evidence-Reasoning to classify fruits. SEP 4 Analysing and Interpreting Data: compare sections across groups. Core competencies: Communication and Collaboration (cooperating while observing and classifying specimens); Critical Thinking and Problem Solving (analysing causes and effects).
Attitudes	Social Justice: the learner shares resources equitably (scalpels, hand lenses, fruit portions) while examining, drawing and labelling sections of fruits. Love: the learner prioritises others' interests during group work. PCI Environment and Technology: appreciates technology in managing the environment. PCI Online Safety: uses only the teacher-cached animation and explores safe internet sites.
Key Inquiry Question	KICD Key Inquiry Questions: (1) What is the significance of reproduction in plants? (2) How does reproduction occur in plants? Lesson focus: What does the inside of a fruit tell us about how seeds are made, and why does a lone tree full of flowers sometimes make no fruit?
Purpose in Storyline	Lessons 1-4 gathered evidence on vegetative propagation, floral structure, and mechanisms that hinder self-pollination and fertilisation, and on double fertilisation. Lesson 5 shows what fertilisation produces: seeds and fruits with distinct internal arrangements. It explains why a lone pawpaw or avocado with many flowers may fail to fruit (no fertilisation, so no seeds or fruit), while cassava and sweet potato cuttings bypass flowers entirely. It raises the next question: Is sexual reproduction better than asexual reproduction for a plant's survival and for farming?
Safety Notes	RAZOR/SCALPEL SAFETY PROTOCOL (explicit): (1) Only one designated 'Cutter' per group of 4-5 handles the blade; the teacher issues blades and counts them out and back in. (2) Always cut on a dissecting tray or board, never in the hand; the fruit is held flat with the fingers well behind the cutting line, cutting away from the body. (3) Blades are carried only in the tray, never walked around the room. (4) Stay seated while cutting; no passing blades hand to hand, so place the blade down on the tray for the next learner. (5) Cover the blade edge with a cork or folded card when not in use. (6) Any cut: stop, raise a hand, and the teacher applies first aid from the kit. (7) Wash hands after handling fruits. Hand lenses are held close to the eye and not pointed at the sun. Do not taste specimens in the lab. Online safety: animation is pre-cached, so no live browsing.

B. LESSON OVERVIEW

Learners begin by recalling the phenomenon: a lone avocado or pawpaw covered in flowers but with no fruit, while cassava cuttings grow with no flower. They predict what is inside a fruit and where seeds are attached, then section tomato, orange, bean pod, passion fruit and cucumber in groups of 4-5 under strict blade safety rules. Using Predict-Observe-Explain, Think-Pair-Share and Claim-Evidence-Reasoning, learners draw, label and classify placentation (axile in tomato and orange, marginal in bean pod, parietal in passion fruit and cucumber; free-central is discussed using a cached image or an available example such as a *Dianthus* or primrose), and connect the seeds they find to fertilised ovules. They add this evidence to the Driving Question Board and revise the class model, ending with a question on whether sexual reproduction is better than asexual reproduction.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Analyzing structure with linear inequalities: fruits Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  PDF: Kno 100021 Passion Fruit Cultivation Source: practical_action  Search ARES for similar readings	Learners look at a teacher-drawn or projected image of a lone avocado tree full of flowers with no fruit next to a cassava cutting sprouting. In pairs they individually sketch in their notebooks what they think the inside of a pawpaw, tomato or passion fruit looks like and where the seeds are attached. They write a one-sentence prediction: 'I think seeds are attached to ____ because ____.' They also predict which part of the flower becomes the fruit. This links back to the phenomenon: why does a flowering tree make no fruit unless a partner tree is near?	Say: 'Remember the lone avocado tree covered in flowers with no fruit. Today we open fruits to see what fertilisation actually produces.' Ask: 'Think of the last time you cut a tomato or passion fruit for a meal. Where were the seeds sitting?' Give WAIT TIME of 10-15 seconds, silently counting. Cold-call 4 learners from different parts of the room (use a class register, not volunteers) to share predictions. Record 3-4 different predictions on the board without judging them. Ask 'Who predicted something different from Wanjiku?' Remind them of recall from Lesson 4: 'What two fusions happen in double fertilisation?' Cold-call 2 learners.	Predict-Observe-Explain (POE): predictions surface prior ideas, such as seeds floating freely inside fruits, so learners are invested in testing them against real sections. Committing to a prediction before cutting makes the later evidence more meaningful and exposes misconceptions.	Check notebooks as you circulate: look for (a) a sketch showing seeds attached somewhere inside the fruit, (b) a written prediction with a reason, (c) mention of ovary or flower part becoming the fruit. Tally on a clipboard how many learners predict seeds attached to a central axis, the wall, or are unattached. Learners who leave the prediction blank get a prompt: 'Which fruit have you cut at home?'
Observe Phase  VIDEO: Bouncing fruit collision example Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  PDF: Kno 100272 Passion Fruit Jam Source: practical_action  Search ARES for similar readings	Safety briefing first (3 min): learners repeat back the 7 blade rules. In groups of 4-5 with assigned roles (Cutter, Holder of tray/specimen, Recorder, Lens Observer/Timekeeper, rotating each fruit so everyone gets a role), they section fruits using shared equipment: tomato and orange (transverse section), bean or green pea pod (longitudinal and transverse), passion fruit and cucumber (transverse). They use hand lenses to locate the ovary wall (pericarp), the placenta, the seeds and their attachment point (funicle), and the chambers (locules). They draw and label each section with a title, magnification note and labels: pericarp, locule, seed, placenta,	Distribute blades by counting: 'One blade per group. I have 10 blades out, and I will count 10 back.' Model one clean cut on a tray in front of the class using a document camera or a student circle. Say: 'Fingers behind the blade, cut away from your body. If you cut yourself, stop and raise your hand.' Circulate and pose questions: 'Where exactly are the seeds attached? Point with your pencil.' 'How many chambers can you count?' 'What does this arrangement look like in the bean pod when you open it along the seam?' Give WAIT TIME of 10-15 seconds after each question. Cold-call at least 6 learners across the groups. After 20 minutes, collect blades and count them before the	Predict-Observe-Explain (Observe step): direct sectioning supplies first-hand evidence that seeds are attached in specific, repeatable ways. Drawing and labelling forces careful attention to structure and ties each seed to the ovary wall, which supports the idea that the fruit is a developed ovary and the seeds are developed ovules.	Labelled dissection diagram check: each learner's drawing must show a title, the correct section type (transverse or longitudinal), at least 4 correct labels (pericarp, locule, seed, placenta/funicle), and a recorded seed attachment point. Observable indicators: correct blade handling (cutting away, on tray), roles rotated, hand lens in use. Mark with a tick-check list: 3/4 labels correct = secure; fewer than 3 = prompt with 'Which part is the wall of the ovary?' Record the count of groups that correctly identify the bean pod attachment along the seam (marginal).

	funicle/axis. They record where the seeds are attached in a table. They share fruit halves so every member and neighbouring group gets to see all fruit types (equitable sharing).	Explain phase. Support equity: check that quieter learners also take the Cutter or Recorder role. Provide pre-cut halves for groups that are slow. For free-central placentation, show a cached photo (such as a Dianthus or primrose fruit) since local fruits may lack it.		
Explain Phase  VIDEO: Rachel Ruysch, Fruit and Insects Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  PDF: Kno 100273 Passion Fruit Juice Source: practical_action  Search ARES for similar readings	Learners use Think-Pair-Share to analyse their evidence: individually (1 min) they decide how seeds are attached in each fruit; in pairs (3 min) they compare and agree; then they share with the class. They then build a Claim-Evidence-Reasoning (CER) statement for each fruit, e.g. Claim: 'Tomato has axile placentation.' Evidence: 'Seeds are attached to a central column where the chamber walls meet, in 2-3 locules.' Reasoning: 'The ovary is divided into chambers by walls that meet at the centre, and ovules attach to this axis, so it is axile.' They complete a classification table: Marginal (bean/pea pod: seeds along one edge/seam of a single-chambered ovary), Axile (tomato, orange: central axis with several chambers), Parietal (passion fruit, cucumber: seeds on the wall of a single chamber), Free-central (central column without partitions, e.g. Dianthus/primrose). They watch a short cached animation on seed and fruit formation after double fertilisation (online safety: only the teacher-cached file) and explain why a lone avocado or pawpaw with flowers but no pollination from another tree gives no seeds and no fruit, whereas a cassava cutting needs no fertilisation. Peers swap tables and review them using a peer review rubric.	Say: 'Turn to your partner and tell them which fruit you would use to explain marginal placentation, and why.' Give WAIT TIME of 10-15 seconds before any pairs speak. Cold-call 5 pairs, then ask a different learner: 'Do you agree? What evidence supports that?' Model one CER aloud for tomato, and push learners with: 'What is your evidence, not just your claim?' Ask: 'If the ovule becomes the seed and the ovary becomes the fruit, what must happen first for the avocado flower?' Cold-call 2 learners. Prompt a link to Lesson 4: 'Which fusion produced the embryo and which produced the endosperm?' Play the cached animation (about 3 minutes), pausing twice with a question: 'What changes in the ovary wall?' Hand out the peer review rubric: correct type, evidence from the section, reasoning links structure to type. Correct misconceptions such as 'the fruit is only the sweet part' by asking 'Is a bean pod a fruit? What does it contain?' (Yes, a legume, a dry fruit.)	Think-Pair-Share lets all 45-50 learners rehearse ideas before public talk, which suits large classes. Claim-Evidence-Reasoning makes learners link observed seed positions to a placentation category, rather than memorising the names. Linking back to double fertilisation and the lone tree phenomenon moves them from description to causal explanation: no fertilisation, so no seeds, so no fruit.	Peer review rubric for the classification table (3 criteria, 0-2 each): correct placentation type per fruit, evidence drawn from the actual section, reasoning that links structure to type. Oral probes: ask 3 randomly selected learners 'Why does the lone pawpaw tree have flowers but no fruit?' Success looks like: 'The flowers were not pollinated or fertilised by another tree, so ovules did not develop into seeds and the ovary did not become a fruit.' Note learners who confuse placentation with pollination for a follow-up. Collect 5 tables for a quick marking sample.

Driving Question Board (DQB) Creation  VIDEO: Add and subtract: pieces of fruit Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  PDF: Kno 100587 Yolanda Morales Growing Passion Fruit In Peru Source: practical_action  Search ARES for similar readings	Learners return to the Driving Question Board. In their groups they write on sticky notes (a) what the sections taught them (e.g. 'Seeds are attached inside the fruit in set patterns, so they come from fertilised ovules'), and (b) a new question (e.g. 'Why do farmers still use cuttings for cassava if seeds give new varieties?' or 'Is it better for a plant to have seeds or cuttings?'). They place their notes next to the original phenomenon (lone avocado/pawpaw vs cassava cutting) and mark which earlier questions have now been answered.	Say: 'Look at our board. Which of our earlier questions can we now answer with evidence from our fruits?' Give WAIT TIME of 10-15 seconds. Cold-call 3 groups to place and justify a note. Highlight a question that leads to Lesson 6, such as 'Is sexual reproduction better than asexual reproduction for a plant's survival and for farming?' Say: 'Notice this question: it compares cassava cuttings and maize seeds. We will gather evidence next time.' Sort questions into themes (fruit and seed development, comparing reproduction types, farming choices).	The DQB makes learning visible and shared. It connects each new piece of evidence to the phenomenon and shows learners that their own questions drive the sequence, which builds ownership and coherence across lessons.	Observable indicators: each group posts at least one 'we learned' note with a piece of evidence (e.g. 'seeds attached to the wall in passion fruit') and one new question. Check that questions go beyond recall (asking 'why' or 'is it better'). Quick tally: number of groups whose question links to sexual versus asexual reproduction.
Model Building Phase  VIDEO: Thomas Hunt Morgan and fruit flies Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Fruit and Seed Dispersal - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Learners revise the class model, a flow diagram of the phenomenon, adding the new link: flower, pollination from a compatible partner plant, double fertilisation, ovule becomes seed, ovary becomes fruit, with placentation showing where seeds attach. They also keep the parallel branch: vegetative propagation (cassava, sugarcane, sweet potato cuttings) bypasses flowers, fertilisation, and seeds. Each learner adds one labelled annotation to their own copy of the model (using colour to show changes made today) and writes a short exit ticket. Learners clean up, return blades, and wash hands.	Say: 'Using your fruit evidence, show on the model what happens to the ovule and the ovary after fertilisation. Where would the lone avocado tree's model stop?' Give WAIT TIME of 10-15 seconds. Cold-call 4 learners for oral probes while circulating: 'How does your model explain why the pawpaw tree needs a partner?' 'What would the model look like for a sweet potato cutting?' Redraw one agreed class model on the board. Hand out exit tickets. Count blades back in and confirm that all are accounted for before dismissal.	Cumulative model revision consolidates evidence from Lessons 1-5 into one explanatory model. Using a colour code for new additions shows how understanding grew and exposes gaps, such as the unanswered question of which route is better for survival and farming.	Exit ticket (3 items): (1) Which flower part becomes the fruit and which becomes the seed? (2) Name the placentation type of tomato and of passion fruit and give one piece of evidence. (3) Explain in one sentence why a lone flowering avocado tree may bear no fruit. Success criteria: correct ovary-to-fruit and ovule-to-seed pairing, correct types with evidence, and reference to missing pollination or fertilisation. Sort tickets into 'secure', 'partly secure', 'needs reteach' for the next lesson's starter.

D. TEACHER REFLECTION

After the lesson, consider: Did every learner get a turn as Cutter, and were all blades counted in and out with no injuries? Which placentation types caused the most

confusion (often axile vs free-central, and parietal vs marginal), and would extra cached images help? Did learners connect fruit structure to fertilisation and to the lone-tree phenomenon, or did they stop at naming types? How well did Think-Pair-Share work in a class of 45-50, and did the peer review rubric produce useful feedback? Were the fruits fresh and sufficient (adjust for seasonal availability of passion fruit or peas)? Which exit-ticket responses suggest a reteach before Lesson 6, which compares sexual and asexual reproduction for survival and farming?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	What did you observe when you sectioned the tomato, orange, bean pod, passion fruit and cucumber? Describe where the seeds were attached and how many chambers each fruit had.
What did I learn?	What did you learn about how fruits and seeds develop after fertilisation, and how do marginal, axile, parietal and free-central placentation differ?
How does this explain the phenomenon?	Using Claim-Evidence-Reasoning, explain why a lone avocado or pawpaw tree full of flowers may not set fruit, while a cassava cutting grows into a new plant without any seed. What new question do you now have about sexual versus asexual reproduction in farming?

LESSON 6 (80 minutes (double lesson)): Variation or Copies? Debating the Trade-offs of Sexual and Asexual Reproduction in Kenyan Crops

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners weigh the advantages and disadvantages of sexual and asexual reproduction using case cards and a comparative chart, so that they can explain why cassava, sugarcane and sweet potato are planted from cuttings while maize, beans, pawpaw and avocado depend on seeds and cross-pollination.
Knowledge	Sexual reproduction (flowers, pollination, fertilisation, seeds) produces offspring that differ genetically from their parents, so there is variation. Asexual reproduction (cuttings, tubers, suckers, runners) produces genetically identical offspring (clones) quickly and without a partner. Each strategy has advantages and disadvantages in farming and in survival.
Skills	NGSS SEP 7 Engaging in Argument from Evidence: learners build and defend claims using case cards. SEP 4 Analysing and Interpreting Data: learners compare cases in a chart. SEP 2 Developing and Using Models: learners revise the class model. Communication and Collaboration: learners cooperate in groups of 4-5 to debate. Critical Thinking and Problem Solving: learners analyse causes and effects while brainstorming on the advantages and disadvantages of sexual and asexual reproduction in plants.
Attitudes	Learners acknowledge the importance of sexual and asexual reproduction in plants. Social Justice: learners share case cards and resources equitably. Love: learners prioritise others' interests by giving each group member a turn to speak and listening to opposing views.
Key Inquiry Question	1. What is the significance of reproduction in plants? 2. How does reproduction occur in plants?
Purpose in Storyline	Lessons 1-5 gave evidence on vegetative propagation, floral structure, pollination barriers, double fertilisation, and seed and fruit development. Lesson 6 turns those findings into an explanation of WHY a plant uses one strategy or the other, by weighing trade-offs of variation versus rapid identical copies. This moves the class closer to answering the driving question and raises the Lesson 7 question: How can we synthesize our findings to explain our local crop phenomenon completely?
Safety Notes	This lesson uses paper materials only, with no blades. Remind learners that the dissecting blade and razor protocols from Lessons 2 and 5 still apply to any later practical: blades are handled only by the designated group member, always cut away from the body, and are passed handle-first and returned to the teacher's tray. Cached animations are played only from the teacher's device (Online Safety: use only pre-checked safe sites). Learners do not taste any plant material. Keep walkways clear in crowded rooms (45-50 learners).

B. LESSON OVERVIEW

Learners begin by predicting what would happen to a field of cassava, or of maize, if a new disease arrived. Using a comparative chart worksheet and case cards (banana wilt in Kisii and Meru banana clones versus variable maize or bean seed crops, cassava mosaic disease, sugarcane in Kisumu and Nyando, sweet potato vines, a lone pawpaw or avocado tree without a partner), groups gather evidence on advantages and disadvantages of each strategy. They then hold a structured debate, explain the phenomenon using Claim-Evidence-Reasoning, add new questions to the Driving Question Board, and revise the class model. The class ends by recognising that neither strategy is better in all conditions and that the trade-off depends on the environment.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Sexual reproduction and genetic variation Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Understand: sexual reproduction and genetic variation Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners individually write a prediction in their notebooks: 'A new fungal disease reaches two farms near Kakamega. Farm A has cassava grown from cuttings of one plant. Farm B has maize grown from seeds from many different plants. What will happen on each farm and why?' They draw a quick sketch of each field, marking healthy and diseased plants. They then share with a partner and record any difference in predictions. Learners link this to the phenomenon: cuttings give identical new plants, seeds give different ones.	Display the two-farm scenario on the board. Say: 'Before we look at any evidence, I want your honest prediction. There are no wrong predictions today.' Give WAIT TIME of 10-15 seconds of silent thinking before writing starts. Walk around and note 3 contrasting predictions. Cold-call 4 learners: 'Tell us what you predicted for Farm A and what you think is the reason.' Say: 'Recall our phenomenon: a cassava cutting becomes a whole plant with no flower, but a lone avocado tree full of flowers sets no fruit. Keep that puzzle in mind.' Record 3 predictions on the board without judging.	Predict-Observe-Explain (POE): Predicting first exposes learners' ideas about identical versus varied offspring, so the case evidence that follows is a test of their own thinking, which builds ownership of the later explanation.	Check notebooks for a written prediction with a reason. Indicators: the learner mentions that plants from cuttings are the same as the parent (or that seeds give differences), and links this to disease spread. Tally on the board how many predict that Farm A suffers more. Note learners who give no reason for follow-up questions.
Observe Phase  VIDEO: Sexual and asexual reproduction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Asexual vs. Sexual Reproduction Practice Source: CK-12  Search ARES for similar readings	In groups of 4-5, learners receive a comparative chart worksheet (columns: Asexual reproduction Sexual reproduction; rows: parents needed, speed, offspring genes, energy and time, dispersal, disease and environmental change, farmer benefit) and a set of 6 case cards: (1) Banana wilt: suckers from one infected plant spread disease through a shamba in Kisii because all plants are identical and equally susceptible. (2) Maize seed variation: in a drought-hit field in Machakos, some seed-grown plants survive because of variation. (3) Cassava cuttings: a farmer in Busia gets a high-yield sweet variety copied fast and cheaply. (4) Sugarcane setts in Kisumu give uniform sugar content and fast harvest. (5) Lone avocado or pawpaw tree has abundant flowers but no fruit without a pollination partner. (6)	Before distributing, say: 'Each of you has a role. The reader reads aloud, the recorder writes, the checker asks, Where in the card is that evidence? and the reporter will share.' Because 45-50 learners are in the class, prepare about 10 sets of cards (photocopied or handwritten) and rotate roles each card to share work equitably. Circulate and ask each group: 'Which card shows an advantage of cuttings? Which shows a disadvantage?' Wait 10-15 seconds after each question before prompting. Cold-call 3 groups to give one row of their chart. Use the cached animation (teacher's device only, 3 minutes) on cross-pollination and seed formation to remind learners how variation arises through meiosis and fertilisation, then pause and ask: 'Where in this animation does the new combination of genes	Evidence-based comparison using a structured chart: sorting case data into two columns lets learners see patterns (speed and uniformity versus variation and resilience) rather than isolated facts. Predict-Observe-Explain continues, as learners compare chart findings with their predictions.	Check charts for at least 5 completed rows with evidence quoted from specific cards. Indicators: learners correctly label banana wilt as a disadvantage of cloning, cassava and sugarcane as advantages of speed and uniformity, and maize variation as an advantage for survival under changing conditions. Ask groups to circle one cell where they are unsure. Use the checker role's questions as a peer check.

	Passion fruit seeds give varied fruit quality, so farmers choose grafts or select seed carefully. Each learner takes one card role (reader, recorder, checker, reporter) and the group fills the chart with evidence from each card. Groups also add one more local example from their home farms.	come from?'		
Explain Phase  VIDEO: Asexual and sexual reproduction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Understand: asexual and sexual reproduction Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups prepare for a structured debate on the motion: 'A Kenyan farmer should plant only cuttings, never seeds.' Half the groups argue for, half against, using the Claim-Evidence-Reasoning (CER) framework on a strip of paper: Claim, Evidence from case cards, Reasoning that links to genetic variation or identical copies. Two rounds of 2-minute speeches are followed by rebuttals. Learners then switch sides for 3 minutes (Think-Pair-Share) to state the strongest point of the opposing side. Finally, each learner writes 3 sentences explaining why cassava is propagated from cuttings while a lone pawpaw or avocado tree needs a partner, linking to variation, pollination barriers and double fertilisation from earlier lessons.	Introduce CER: 'A claim without evidence is just an opinion. Every point you make needs a card number.' Assign sides by group number. Set a visible timer. Say: 'Listen so that you can repeat your opponent's best point before answering it.' After each speech give WAIT TIME of 10-15 seconds before rebuttals. Cold-call 5 learners from different groups: 'What evidence from the cards supports that claim?' During switch-sides, say: 'Turn to your partner and give the strongest argument for the other side. I will ask 3 pairs.' Highlight the key idea on the board: Sexual reproduction = variation, a partner, and slower spread; Asexual = identical copies, no partner, rapid colonisation, and shared risk. Ask: 'So why can a stem cutting grow with no flower while a lone avocado tree cannot set fruit?' Guide the class to see that fruit needs fertilisation and often cross-pollination, while cuttings regrow from existing tissue.	Claim-Evidence-Reasoning (CER) in a debate, supported by Think-Pair-Share: structured argument forces learners to justify trade-offs with evidence, and arguing the opposite side strengthens understanding that neither strategy is universally better.	Use a peer-review rubric (1-3 points each) for Claim clarity, Evidence from at least 2 cards, and Reasoning linking to variation or identical copies. Indicators of mastery: the learner states that identical plants are equally vulnerable to one disease, that variation gives some individuals a chance of surviving, and that seeds help dispersal to new areas. Collect the 3-sentence explanations as an exit ticket: 'Name one advantage and one disadvantage of each strategy using a Kenyan crop.' Sort into Got it / Almost / Not yet piles.
Driving Question Board (DQB) Creation  VIDEO: Sexual reproduction and	Learners review the class DQB. Each group writes one new question or answers one existing question on sticky notes or paper strips, for example: 'Could a farmer in Kericho combine both strategies, for example grafting?'	Say: 'Look at our board. Which questions can we now answer, using today's evidence?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners to move a question and justify it using a case card. Ask: 'What new question	Driving Question Board (DQB) tracking: moving questions from 'wondering' to 'answered' makes growth in understanding visible and shows learners that new evidence generates new questions, keeping the	Observe whether notes move from fact-recall (What is a clone?) to explanatory or application questions (Why does a clone suffer more in an epidemic?). Indicator: at least half of the groups pose a how or why

genetic variation Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Understand: sexual reproduction and genetic variation Source: Khan Academy (English - US curriculum) Search ARES for similar readings	'Why do pawpaw trees sometimes have separate male and female plants?' 'Would a disease also wipe out a field of sweet potato vines?' Learners move answered questions to the 'Answered' column with evidence and place new questions in 'Still wondering'.	does today's debate raise?' Select 2 new questions to read aloud. Point to the one that links to the next lesson: 'How can we synthesize our findings to explain our local crop phenomenon completely?' Make sure all groups, including quieter ones, post at least one note so that all voices are shared equitably.	phenomenon at the centre.	question that links two lessons. Record which questions remain unanswered for Lesson 7.
Model Building Phase VIDEO: Sexual reproduction and genetic variation Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Understand: sexual reproduction and genetic variation Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners revise the class model for the phenomenon (a large poster with two pathways: 'Cutting/vegetative' and 'Flower/seed'). They add: (a) on the vegetative pathway, identical offspring, rapid spread, no partner needed, vulnerability to one disease (banana wilt); (b) on the seed pathway, flowers, cross-pollination, double fertilisation, variation, and resilience to disease or drought. Each learner also revises a personal model in the notebook with a different colour pen and writes one sentence beginning, 'I changed my model because...'	Say: 'Our model must explain both halves of the puzzle: the cutting that grows without a flower and the lone tree with flowers but no fruit.' Invite 4 volunteers (or cold-call 4) to add one element each and justify it with evidence. Ask probing questions: 'What does the arrow from flower to seed represent?' 'Where does variation enter?' Wait 10-15 seconds before accepting answers. Circulate to check personal models and ask 5 learners orally: 'What did you add today, and which evidence made you add it?' Close by saying: 'Next lesson we pull all the pieces together to fully explain our crops.' Distribute exit tickets: 'Which strategy would you advise a farmer in your village to use for (a) cassava and (b) beans, and why?'	Cumulative model revision: adding new components and colour-coded changes to a shared model helps learners integrate evidence from all lessons into one coherent explanation of the phenomenon and see their own conceptual change.	Oral probes during model building: listen for correct use of the terms variation, clone, pollination, and fertilisation. Indicators: the revised model shows identical offspring from cuttings, new gene combinations from seeds, and a reason (pollination partner or disease risk) linked to the lone tree or the banana field. Exit ticket answers reveal whether learners can apply trade-offs to specific crops. Mark labelled diagram checks in notebooks for accuracy.

D. TEACHER REFLECTION

After the lesson, consider: Did learners support their claims with specific evidence from the case cards, or did they rely on opinion? Which misconceptions surfaced (for example, that asexual reproduction is always 'better' because it is faster, or that seeds always come from one plant)? Did large class size limit participation, and would role cards or smaller debate teams help? Which groups needed extra support with CER? Check whether the cached animation was clear and whether the case cards used familiar Kenyan crops. Note unanswered DQB questions to shape Lesson 7 and decide whether any learners need revision on pollination barriers or double

fertilisation before the synthesis task.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed from the case cards that plants grown from cuttings, suckers or setts (cassava, sugarcane, sweet potato, banana) are identical to the parent and spread quickly, but one disease like banana wilt can affect all of them. Seed-grown crops such as maize, beans and passion fruit vary from plant to plant, and fruiting trees like a lone pawpaw or avocado may not set fruit without a partner tree.
What did I learn?	We learned that asexual reproduction needs only one parent and gives fast, identical offspring that are useful for colonising an area and keeping good traits, while sexual reproduction needs pollination and fertilisation and gives genetic variation that helps some offspring survive disease and changing conditions. Each strategy has advantages and disadvantages.
How does this explain the phenomenon?	We can explain that cassava, sugarcane and sweet potato can grow from cuttings with no seed because new plants regrow from parent tissue and are clones, while maize, beans, pawpaw and avocado produce fruit and seeds through flowers, pollination and fertilisation, which creates variation and often needs another plant. A lone tree full of flowers may not fruit because it lacks a pollen source, whereas a stem piece needs no flower or partner to start a new plant.

LESSON 7 (80 minutes (double lesson)): Putting the Evidence Together: Why Cassava Cuttings Need No Seeds but Lone Pawpaw and Avocado Trees Need a Partner

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students integrate evidence from Lessons 1-6 to explain why cassava, sugarcane and sweet potato can be grown from cuttings with no seeds or flowers, while maize, beans and lone pawpaw or avocado trees depend on flowers, pollination, double fertilisation and seed or fruit formation. They revise their Day 1 models and present group explanatory models.
Knowledge	By the end of the lesson the learner should be able to: (a) describe asexual reproduction in plants (vegetative propagation by cuttings, runners, rhizomes, tubers); (b) identify flowers based on their floral parts; (c) illustrate mechanisms and features that hinder self-pollination and self-fertilisation; (d) describe the process of double fertilisation in flowering plants; (e) acknowledge the importance of sexual and asexual reproduction in plants.
Skills	NGSS SEP: Developing and Using Models (revise and present a group model); Constructing Explanations (Claim-Evidence-Reasoning); Engaging in Argument from Evidence (peer review); Obtaining, Evaluating and Communicating Information. Practical skills: dissecting and labelling a flower, classifying a fruit by placentation and type.
Attitudes	Core competencies: Communication and Collaboration (cooperating with peers while observing and classifying flower specimens); Critical Thinking and Problem Solving (analysing causes and effects while weighing the advantages and disadvantages of sexual and asexual reproduction). Values: Social Justice (sharing dissecting kits, hand lenses and specimens equitably); Love (prioritising others' interests when using models and charts). PCIs: Environment and Technology (appreciating technology in managing the environment) and Online Safety (using only teacher-approved, pre-downloaded animations).
Key Inquiry Question	1. What is the significance of reproduction in plants? 2. How does reproduction occur in plants?
Purpose in Storyline	Lesson 7 is the integration point of the evidence-gathering sequence. Students bring together vegetative structures (Lessons 1-2), floral adaptations and barriers to self-pollination (Lessons 3-5) and double fertilisation, seed and fruit development (Lesson 6) to answer the driving question. It ends with a new need: How do we format our complete scientific explanation for final assessment?
Safety Notes	Scalpels, razor blades and mounted dissecting blades: cut away from the body and on a tile or board, never in the hand. Only one cutter per group of 4-5, handled by the designated 'cutter' while others observe. Blades are carried closed or in a tray and counted in and out by the teacher. Report any cut immediately and cover it with a plaster from the first-aid kit. Wash hands after handling plant material, because some flowers and sap, such as cassava latex, irritate skin or are toxic when raw; no tasting of specimens. Animations: use only pre-downloaded files, with no live browsing.

B. LESSON OVERVIEW

Students start by recalling the puzzle: a piece of cassava stem becomes a whole plant with no flower, yet a lone pawpaw or avocado tree covered in flowers gives no fruit. In a Predict-Observe-Explain flower dissection (hibiscus, with tomato or passion fruit flowers as available), they predict which floral features would hinder self-pollination, then observe and label them. Think-Pair-Share on placentation of a cut tomato and passion fruit follows, and a CER task classifies fruits. Groups then revise their Day 1 models on chart paper, post sticky-note answers on the Driving Question Board (DQB), and present a model linking vegetative structures, floral adaptations and double fertilisation. Exit tickets surface the remaining question about how to write the full scientific explanation.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Sexual reproduction and genetic variation Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Understand: sexual reproduction and genetic variation Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Students sit in their groups of 4-5 with their Day 1 model of the phenomenon. Individually they write a quick prediction in their notebooks: 'Which parts of a flower could stop a lone pawpaw or avocado tree from setting fruit, and why does a cassava cutting not need these parts?' They sketch where they expect the anthers, stigma and ovary to be in a hibiscus flower and predict what a cut-open ovary will show. Then they share with their group and record one group prediction on a sticky note.	Display the phenomenon on the board with a drawing of a cassava cutting beside a flowering lone pawpaw tree. Say: 'Before we touch any specimen, look back at your Day 1 model. What would you change today, based on what we have learned?' Give WAIT TIME of 10-15 seconds, silently counting, before taking answers. Cold-call 4 students from different groups: 'Njeri, what did your group predict about why the lone pawpaw has flowers but no fruit?' Ask follow-up: 'What evidence from our earlier lessons supports that?' Circulate and note common misconceptions (for example, 'flowers are only for decoration' or 'a cutting is the same as a seed'). Hand out specimens with a designated 'cutter', 'recorder', 'reporter', 'materials manager' and 'safety monitor' in each group, rotating the roles to share duties fairly. Briefly restate the blade safety rules before kits are handed out.	Predict-Observe-Explain (POE): predicting first makes students commit to ideas that they can later test against evidence. This exposes preconceptions and builds the need to understand, because they want to know whether they were right. It connects directly to the anchoring phenomenon by asking them to forecast which structures explain lone-tree failure and seedless propagation.	Observable indicators: (1) each student has a written prediction with a reason in their notebook; (2) at least 4 cold-called students mention floral parts (stigma, anther, ovary) or a barrier such as different maturation times or separate male and female trees; (3) the teacher records on a checklist the groups still holding the misconception that cuttings need flowers or seeds.
Observe Phase  VIDEO: Sexual and asexual reproduction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Asexual vs. Sexual Reproduction Practice	Groups dissect a hibiscus flower (or a tomato or passion fruit flower if available) following a teacher demonstration. Using hand lenses they locate and draw the sepals, petals, stamens (anther and filament), stigma, style and ovary, then cut the ovary across to see ovules. They note features that hinder self-pollination, such as stigma and anthers at different heights (herkogamy) or anthers releasing pollen before the stigma is receptive (protandry). They compare with a labelled chart of a pawpaw with separate male and	Demonstrate one safe cut first: 'Blade away from your body, flower on the tile, only the cutter touches the blade, and everyone else keeps their hands behind the line.' Check that every group has a counted blade, and collect the blades in a tray at the end. Circulate and probe: 'Show me where pollen would land. How far is it from the anthers? What does that tell you about self-pollination?' Ask the maize question: 'Where is the male part on maize and where is the female part? How does a maize plant get pollen onto silk?'	Predict-Observe-Explain (POE): observation of real specimens gives concrete evidence against which predictions are checked. Dissection plus animation links the visible structure (anther-stigma distance, ovule number) to the unseen process (double fertilisation), building a mechanism for why flowering plants need a partner plant and why cuttings do not.	Labelled dissection diagram check: the teacher reviews one diagram per group for correct labels (stigma, style, ovary, ovule, anther, filament, petal, sepal) and at least one correctly identified barrier to self-pollination. Peer review rubric for the floral classification table: partners tick (1) ovary position correct, (2) symmetry correct, (3) male and female parts correctly identified, and (4) one clear note on the pollination mechanism. Safety indicator: all blades returned and counted.

Source: CK-12 Search ARES for similar readings	female trees (dioecious) and maize with male tassels and female cobs (monoecious). A 3-minute pre-downloaded animation of double fertilisation is played twice for the class. Each group completes a labelled dissection diagram and a floral classification table (ovary position, symmetry, male and female parts present).	Play the cached animation, pause at pollen tube growth, and ask: 'What two things fuse here, and what does each become?' WAIT TIME of 10-15 seconds, then cold-call 3 groups. Use a tablet or laptop only for the stored files and remind students that online safety means only teacher-approved materials are used. For the large class, run the dissection in staggered stations: 2 groups dissect while others complete the classification table, then swap after 8 minutes.		
Explain Phase  VIDEO: Asexual and sexual reproduction Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Understand: sexual and asexual reproduction Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Think: students silently examine a cut tomato (or passion fruit) and a bean pod showing seed attachment, and sketch where the seeds attach (axile in tomato, marginal in bean, parietal in passion fruit). Pair: they discuss with a partner how the placentation relates to the ovary structure and how it differs between fruits. Share: pairs report to the class. Then each pair writes a CER statement for classifying one fruit: Claim (for example, 'The tomato is a fleshy simple fruit (berry) with axile placentation'); Evidence (observations of seed attachment, fleshy pericarp, number of locules); Reasoning (fruit develops from the fertilised ovary, ovules become seeds, and placentation is determined by how ovules attach in the ovary). They also complete a quick comparison of advantages and disadvantages of cuttings (cassava, sugarcane, sweet potato vine) and seeds (maize, beans).	Give the instruction: 'Think alone for 2 minutes. Draw what you see before you talk.' Then: 'Turn to your partner and explain how the seeds are attached and why.' Allow WAIT TIME of 10-15 seconds before calling on pairs and cold-call 5 pairs, including from the back rows. Probe with: 'What part of the flower became this fruit? What happened to the ovules?' and 'Why does a cassava cutting produce plants identical to the parent, while maize seeds produce varied plants?' Model a CER on the board using a pawpaw example, then ask pairs to write their own. Challenge: 'If your claim is that cross-pollination is needed, what evidence from the lone avocado would convince a farmer in Kiambu?' Draw out the advantages of asexual reproduction (fast, identical, no pollinator needed) and disadvantages (no variation, disease spreads), contrasted with sexual reproduction (variation, dispersal, adaptation to changing conditions). Ask for the significance of reproduction: 'How does each method help a farmer in	Think-Pair-Share builds understanding by giving individual thinking time, then rehearsal with a partner, before public sharing; this lets quieter students contribute in a class of 45-50. Claim-Evidence-Reasoning forces students to tie fruit classification to observations and to the underlying process of ovary-to-fruit development, linking the reproductive mechanisms back to why fruit forms only after pollination and fertilisation in the phenomenon.	Indicators: (1) the student sketch shows seeds attached correctly for at least two fruit types; (2) a CER statement includes an observation as evidence and links it to ovary and ovule development as reasoning; (3) in the cold-call, at least 4 of 5 pairs correctly connect fertilisation to fruit formation; (4) pairs identify at least one advantage and one disadvantage each for asexual and sexual reproduction. The teacher scores CERs 0-3 on a quick tally.

		Kisumu or Kericho keep food supply going?		
Driving Question Board (DQB) Creation  VIDEO: Sexual reproduction and genetic variation Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Asexual Reproduction Practice Source: CK-12  Search ARES for similar readings	Students go to the Driving Question Board with sticky notes. Each group posts (a) one answer to an earlier question now resolved with evidence (for example, 'Why does the lone pawpaw have no fruit? Pollen from another tree is needed and the male and female flowers are on separate trees') and (b) one new or remaining question (for example, 'How do we write all this into one clear explanation?'). Questions are sorted into themes: vegetative propagation, pollination barriers, double fertilisation, and fruit and seed formation. Students take a gallery walk and add a star to the answer they find most convincing.	Say: 'Choose a question from the board that you can now answer using evidence from our dissection or our earlier lessons. Put your evidence on the note, not just your opinion.' Cold-call 3 students to read out their note and ask the class: 'Does that evidence support the claim? Who can add to it?' After WAIT TIME of 10-15 seconds, invite responses. Group sticky notes into themes and point out unresolved questions. Highlight the new question: 'Scientists must communicate explanations clearly. What do we still need to know to write a complete explanation for assessment?' Ensure each group, not only the dominant voices, places a note, so that contributions are shared equitably.	The Driving Question Board makes the class's collective thinking visible and traces growing understanding across lessons. Matching answers to the original questions shows students how evidence resolves puzzles, and the leftover questions create purpose for the next lesson, keeping the phenomenon of seedless cuttings versus partner-dependent fruiting at the centre.	Indicators: each group posts at least one evidence-based answer and one new question; at least 3 questions have been moved from 'unanswered' to 'answered'; the teacher notes on a checklist which themes have the fewest answered questions to target in the next lesson.
Model Building Phase  VIDEO: Sexual reproduction and genetic variation Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Understand: sexual reproduction and genetic variation Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups take their Day 1 model and a sheet of chart paper. They revise it into a group explanatory model with two linked pathways: (1) asexual pathway - cassava, sugarcane and sweet potato cuttings, runners, rhizomes and tubers produce genetically identical plants without flowers or fertilisation; (2) sexual pathway - flower structure, barriers to self-pollination (dioecy in pawpaw, protandry, herkogamy, self-incompatibility in avocado), cross-pollination by insects or wind, double fertilisation (one sperm with the egg to form the zygote, the other with the polar nuclei to form endosperm), and seed and fruit development. They colour-code what changed from Day 1 and write a 'because' statement	Instruct: 'Use a different colour marker to show what you have changed from your Day 1 model, and label each change with the evidence you used.' Circulate and ask oral probes: 'Where in your model does the lone avocado fail? What would you add so a farmer in Nakuru gets fruit?' 'Where does the endosperm in maize come from and why does ugali depend on it?' Allow WAIT TIME of 10-15 seconds after each probe. Choose 3 groups to present, making sure different groups are selected than in previous lessons; cold-call 3 audience members to ask one question each using the sentence starter 'Your model shows... but what about...?' Summarise: both strategies help plants survive and farmers produce food: cuttings	Cumulative model revision: students use new evidence to modify their initial ideas, which develops the SEP of Developing and Using Models. Presenting and questioning models allows argument from evidence and shows how different structures (stems, flowers, ovules) combine to explain the Kenyan farm phenomenon. It connects the sub-strand outcomes into one coherent causal explanation.	Oral probes during model building: students can state why a cutting needs no flower and why a lone pawpaw or avocado often fails to fruit. Model checklist: (1) asexual and sexual pathways both shown; (2) a barrier to self-pollination named; (3) double fertilisation correct with two products (zygote and endosperm); (4) at least one revision marked from Day 1. Exit ticket (3 minutes): 'Explain in 3 sentences why a farmer can grow cassava from a cutting but needs two trees for fruit from pawpaw. Write one question about how to present your full explanation.' Teacher sorts tickets into secure, partial and not yet.

	answering the driving question. Two or three groups present in 2 minutes each in a gallery format while others ask questions.	give speed and uniformity, while seeds give variation and spread. Close by posing: 'We now have the pieces. How do we format our complete scientific explanation for final assessment?' Collect exit tickets at the door.		
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D. TEACHER REFLECTION

After the lesson ask yourself: Which groups could link cuttings, floral barriers and double fertilisation in one causal chain, and which still treated them as separate topics? Did the staggered dissection stations keep all 45-50 students engaged and every blade accounted for? Which students did not speak during the cold-calls, and how will you include them next time? Which misconceptions remain (for example, that a cutting needs a flower, that pollination and fertilisation are the same event, or that every flower becomes a fruit)? Check the exit tickets to decide what to teach in Lesson 8 on structuring a Claim-Evidence-Reasoning scientific explanation, and whether the cached animations and shared equipment worked well under the school's limits.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What we observed: a hibiscus (or tomato or passion fruit) flower dissected to show stigma, style, ovary, ovules, anthers and filaments; the distance between anthers and stigma; fruits cut open to show placentation; the animation of double fertilisation; and cassava, sugarcane and sweet potato cuttings that grow into new plants without seeds.
What did I learn?	What we learned: asexual reproduction by cuttings and other vegetative parts gives identical plants quickly without flowers or pollination; sexual reproduction needs flowers, pollination (often cross-pollination because of barriers such as separate male and female trees, different timing and self-incompatibility), double fertilisation, and then seed and fruit development; and both methods are important for farming.
How does this explain the phenomenon?	How this explains the phenomenon: a cassava cutting grows from stem tissue of the parent, so no flower or partner is needed. A lone pawpaw or avocado tree has flowers but barriers to self-pollination prevent fertilisation, so without pollen from another tree the ovules are not fertilised and no fruit or seed forms. Maize and beans must come from seeds because they reproduce sexually: pollination and double fertilisation produce the embryo and endosperm inside the seed.

LESSON 8 (80 minutes (double lesson)): Final Explanation: Why Cassava Cuttings Grow Without Seeds but a Lone Pawpaw Needs a Partner Tree

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students synthesise evidence from Lessons 1-7 into a final individual Claim-Evidence-Reasoning (CER) explanation of the anchoring phenomenon, peer review it with a rubric, compare their Lesson 1 and final models, close out the Driving Question Board (DQB) and complete the unit assessment tasks.
Knowledge	By the end of the sub strand, the learner should be able to: a) describe asexual reproduction in plants, b) identify flowers based on their floral parts, c) illustrate mechanisms and features that hinder self-pollination and self-fertilisation in plants, d) describe the process of double fertilisation in flowering plants, e) acknowledge the importance of sexual and asexual reproduction in plants. Key ideas: cuttings of cassava, sugarcane and sweet potato regrow by mitosis from vegetative tissue and give clones; maize, beans, pawpaw and avocado fruit only after pollination, double fertilisation and seed and fruit development; features such as separate sexes, different timing of maturity of anthers and stigma, and self-incompatibility hinder self-pollination and explain why a lone flowering tree may not set fruit.
Skills	NGSS Science and Engineering Practices: Constructing Explanations (CER); Engaging in Argument from Evidence (peer review); Developing and Using Models (L1 vs final comparison); Obtaining, Evaluating and Communicating Information. Core competencies: Communication and Collaboration (cooperating with others while observing and classifying specimens of flowers); Critical Thinking and Problem Solving (analysing causes and effects while brainstorming on the advantages and disadvantages of sexual and asexual reproduction in plants).
Attitudes	Values: Social Justice (shares resources equitably while examining, drawing and labelling sections of fruits); Love (prioritises others' interests while using models and charts to illustrate mechanisms and features that hinder self-pollination and self-fertilisation in plants). PCIs: Environment and Technology (appreciates the use of technology in managing the environment while watching animations illustrating double fertilisation and formation of seeds and fruits); Online Safety (explores safe internet sites while watching animations illustrating double fertilisation and formation of seeds and fruits).
Key Inquiry Question	1. What is the significance of reproduction in plants? 2. How does reproduction occur in plants?
Purpose in Storyline	Final lesson of the storyline. Lessons 1-7 gathered evidence on vegetative propagation, flower structure, features that hinder self-pollination, double fertilisation, and seed and fruit development with placentation. Lesson 8 pulls this together so each learner can answer the Driving Question: Why do some farm plants produce offspring without seeds while others require seeds and a partner tree to make fruit? There is no new question for next lesson because the unit is complete.
Safety Notes	Blade and razor safety (from Lessons 2 and 5) is revisited: no dissection is planned today, but if any specimen is re-examined for evidence, only one trained student per group handles the blade, which stays in a labelled tray, is passed handle-first and is never carried while walking; hand lenses are shared in groups of 4-5. Cached animations are played offline by the teacher; remind learners of Online Safety: only use teacher-approved sites and never open unknown links. Wash hands after handling plant material, and do not taste specimens in the lab.

B. LESSON OVERVIEW

Learners return to the market-garden puzzle: a cassava stem cutting becomes a whole plant with no flower, while a lone pawpaw or avocado in a compound is covered in flowers but sets no fruit. They first predict how their ideas have changed, then use their notebooks, labelled dissection diagrams, fruit classification tables and class

charts as evidence to write an individual CER explanation. Peer review with the rubric, a comparison of the Lesson 1 and final models, and closing of the DQB follow. The lesson ends with the unit assessment tasks and an exit ticket. Pre-downloaded animations and a printed one-page evidence summary per group cope with the large class (45-50) and limited internet. Sensemaking strategies: Predict-Observe-Explain (POE), Think-Pair-Share, CER and DQB model revision.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Sexual and asexual reproduction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Asexual vs. Sexual Reproduction Practice Source: CK-12  Search ARES for similar readings	Learners get an index card showing two scenarios: (a) a cassava stem cutting planted in Kisumu soil that grows roots with no flower, and (b) a lone avocado tree in a Nairobi compound with hundreds of flowers but no fruit. Individually they write a 2-3 sentence prediction: 'What would happen if we planted a second avocado tree 10 m away, and what would happen to a cassava cutting planted beside it?' and say which processes (mitosis, pollination, fertilisation) are involved. They then predict what a stronger final explanation needs to contain, to link back to the phenomenon.	Stand at the door and say: 'Today you are the experts. Before we start, commit to a prediction on your card. Write silently for 3 minutes.' Display the two scenarios on the board with a sketch of a cassava cutting and an avocado tree. After writing, say: 'Turn to your shoulder partner and share your prediction in 30 seconds each.' Then cold-call 4 learners from different groups: 'Wanjiru, what did you predict for the lone avocado? What evidence from our lessons supports that?' WAIT TIME: 10-15 seconds after each question before choosing a respondent. Record 3-4 predictions on the board without evaluating them. Say: 'Keep these predictions. We will test them against all our evidence.'	Predict-Observe-Explain (POE): predicting first makes learners commit to ideas drawn from the whole unit so they can notice later which of their ideas the evidence supports or corrects. Linking the scenarios to the anchoring phenomenon surfaces remaining misconceptions, such as 'flowers always make fruit' or 'cuttings are a type of seed'.	Observe: (1) at least 80% of learners write a prediction that names a mechanism (pollination from another tree, mitosis in the cutting) and not just 'it needs water'; (2) learners use terms such as 'cross-pollination', 'clone', 'self-incompatible'. Note learners who still say 'cuttings need pollen' for the oral probe in the model phase. Entry check: collect 10 cards at random to find common gaps.
Observe Phase  VIDEO: Asexual and sexual reproduction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Understand: sexual and asexual	In groups of 4-5 learners rotate through four evidence stations set around the room (tables or desk clusters), spending about 3 minutes at each. Station A: vegetative propagation charts and photos of cassava, sugarcane and sweet potato cuttings, plus a note on runners, bulbs, rhizomes and tubers (Lesson 1-2). Station B: their labelled flower diagrams and classification tables for hibiscus, maize, passion fruit, bean and tomato flowers, including features that hinder self-pollination	Say: 'Your task is to collect your strongest piece of evidence from each station. You cannot write a claim without evidence.' Circulate with a checklist. Pre-assign a materials manager and a recorder per group to ensure equitable sharing of the hand lenses and charts (Social Justice). Ask at each station: 'Where in your notebook do you see that? How does this help explain why the lone pawpaw does not fruit?' WAIT TIME 10-15 seconds before giving hints. Cold-call at least 6 learners	Evidence review through POE 'Observe': learners re-observe their own data and class artefacts rather than being told answers, and sort them into two causal stories: (1) asexual reproduction: mitosis, clone, no pollen needed, and (2) sexual reproduction: pollination, double fertilisation, seeds and fruit, genetic variation. This pre-organises the evidence for the CER.	Check the evidence logs for at least 4 pieces of evidence (one per station) with correct lesson sources. Observable indicators: learners correctly identify a cutting as producing a clone (Station A); learners point to a feature that hinders self-pollination, such as separate male and female flowers in maize or stigma maturing later in hibiscus (Station B); learners label endosperm and zygote correctly (Station C); learners link ovary to fruit and ovules to seeds (Station D). Labelled diagram

reproduction Source: Khan Academy (English - US curriculum) Search ARES for similar readings	(Lessons 3-5). Station C: the cached double fertilisation animation screenshots and the group summary of zygote and endosperm formation (Lesson 6), shown from a printed sequence. Station D: fruit sections of tomato, passion fruit and bean pod with placentation and fruit type classification (Lesson 7). At each station learners tick off and write one piece of evidence that helps answer the Driving Question in their evidence log and note the lesson where it came from.	across the groups during the rotation. For safety, remind groups: 'Any blades stay in the tray at the front; today we only use hand lenses.' Use a timer and signal each rotation with a bell or clap. For the offline animation, play the cached clip once on the teacher's device, projected, with the sound off, and ask learners to narrate the steps.		check: mark with a tick or a note where labels are missing.
Explain Phase  VIDEO: Sexual and asexual reproduction Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Asexual Reproduction Practice Source: CK-12 Search ARES for similar readings	Individually, learners write their final Claim-Evidence-Reasoning explanation on the CER template (12 minutes). Claim: a 1-2 sentence answer to the Driving Question. Evidence: at least three pieces of evidence from the unit (for example cassava cuttings regrowing from nodes without flowers, maize and beans producing seeds after pollination, the flower diagram of a single hibiscus showing anthers and stigma, double fertilisation animation, placentation in the passion fruit or tomato, the lone avocado with flowers but no fruit). Reasoning: connect the evidence to the science ideas: mitosis produces identical cells so a cutting forms a clone; pollination and fertilisation combine gametes from two parents; features such as different maturity times of anthers and stigma (dichogamy), separate sexes and self-incompatibility hinder self-fertilisation so another tree is needed as the pollen source; double fertilisation forms the zygote and endosperm, then ovules become seeds and ovary becomes fruit. They also include	Say: 'A strong explanation must answer both parts of our puzzle: why the cutting works without seeds and why the lone tree fails to fruit.' Model the CER structure by displaying a partial example on the board about sweet potato vines: 'Claim: ... Evidence: ... Reasoning: ...' and ask: 'What is missing from this reasoning?' WAIT TIME 10-15 seconds. During writing, circulate and use prompts: 'Which observation supports your claim? Why does that evidence matter?' Prioritise support for learners who need it by pairing with a partner and offering sentence starters ('This shows that ... because ...'). For Think-Pair-Share say: 'Pair: read your partner's CER silently, then give one star and one step using the rubric. You have 5 minutes per person.' Cold-call 5 pairs to share one improvement they suggested, asking: 'Kamau, what step did your partner give you and why does it make the reasoning stronger?' Remind learners of Love: 'Give feedback that helps your partner, kindly and specifically.' Collect two CERs on the board to draw out	Claim-Evidence-Reasoning (CER) with Think-Pair-Share: CER forces learners to separate what they know (claim), what they saw (evidence) and why it matters (reasoning), which makes gaps in their thinking visible. Peer review with the rubric turns learners into critics of evidence, strengthening Engaging in Argument from Evidence, and Think-Pair-Share gives every learner talk time in a large class of 45-50.	Use the peer review rubric and a teacher checklist. Observable indicators of success: (1) the claim directly answers 'why some plants reproduce without seeds while others need seeds and a partner tree'; (2) at least three correct pieces of evidence with specific Kenyan examples; (3) reasoning that includes mitosis and clones for cuttings, plus pollination, double fertilisation and self-pollination barriers for seed-bearing plants; (4) correct use of terms (ovary to fruit, ovule to seed). Common errors to note: confusing pollination with fertilisation, saying a cutting is a seed, saying all flowers make fruit by themselves. Sample at least 10 CERs during the circulation and give oral feedback immediately. Peer review rubrics are collected as evidence of the floral classification table review.

	one advantage and one disadvantage of each type of reproduction (for example cuttings give fast, uniform cassava crops but spread disease; seeds give variation that helps adaptation but need pollinators). Then Think-Pair-Share (13 minutes): swap with a partner, review using the peer review rubric (claim answers question; at least 3 pieces of evidence; reasoning connects evidence to claim with correct terms; Kenyan examples), give one 'star' and one 'step', and revise for 3 minutes.	common strong reasoning.		
Driving Question Board (DQB) Creation  VIDEO: Sexual and asexual reproduction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Asexual vs. Sexual Reproduction Practice Source: CK-12  Search ARES for similar readings	The class revisits the DQB created in Lesson 1 (chart paper on the wall with sticky notes). In their groups, learners take 5 minutes to sort the notes into three columns: 'Answered by our evidence', 'Partly answered', and 'New question for future study'. For each answered question, a learner writes the answer and the lesson that supplied the evidence on a new sticky note (for example 'Why does cassava not need seeds? Mitosis in the stem cutting forms roots and shoots, Lessons 1-2'). Groups then add at least one new question that goes beyond the unit, such as 'Can tissue culture produce banana plantlets from a few cells?' or 'How do bees affect passion fruit yields in Kericho?'	Say: 'Our wall started with your questions. Now it is time to see how far we have come.' Hand each group a different colour of marker to track contributions. Ask: 'Which question was hardest to answer and why?' WAIT TIME 10-15 seconds. Cold-call 3 groups to read one answered question and the evidence behind it. Say: 'If a question is only partly answered, what extra evidence would we need?' Because there is no next lesson, record the leftover questions on a 'Future Investigations' list to be kept in the class file for the teacher and for science club projects. Reinforce Environment and Technology by asking how farmers use technology such as tissue culture or improved seed to manage crops.	Driving Question Board: revisiting the learners' own questions shows them how their understanding has grown and makes learning visible and learner-owned. Sorting questions by evidence level builds the habit of linking claims to data, and leftover questions reveal that science keeps generating new inquiry.	Look for: at least 80% of the original DQB questions are moved with an evidence-based answer; sticky-note answers cite a specific lesson and observation; new questions are testable or researchable. Note any questions still unresolved (for example confusion between self-pollination and self-fertilisation) for follow-up feedback.
Model Building Phase  VIDEO: Sexual and asexual reproduction	Learners take out the model they drew in Lesson 1 (a diagram showing how they thought cassava cuttings and lone pawpaw or avocado trees reproduce). In pairs, they draw a final model on a new page in 8 minutes showing two	Say: 'Your first model was your best thinking at the beginning. Compare it with your final model and show me how your thinking has changed.' Walk around to listen and ask oral probes: 'Where does the pollen come from in your	Cumulative model revision: comparing the L1 model with the final model shows conceptual change, and explaining what evidence caused each change builds scientific reasoning and metacognition. The assessment	Oral probes during model building: learners correctly explain that cuttings do not need pollen and give clones, and that the lone tree lacks compatible pollen from another tree. Model rubric indicators: both pathways present;

Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Asexual vs. Sexual Reproduction Practice Source: CK-12 Search ARES for similar readings	pathways: (1) asexual: stem cutting, mitosis, roots and shoots, clone; (2) sexual: flower with anthers and stigma, pollen transferred from another tree (cross-pollination), pollen tube, double fertilisation producing zygote and endosperm, ovule becoming seed, ovary becoming fruit; with annotations for features hindering self-pollination and a note on the advantages and disadvantages of each. They then place the L1 and final models side by side and write three sentences: 'What I changed', 'What evidence made me change it', 'What I still wonder'. Finally, learners complete the unit assessment tasks individually (10 minutes): short-answer items and a labelled diagram from the unit assessment sheet, marked against the rubric. They finish with an exit ticket.	model of the pawpaw? What would happen if you had only one tree? Why does the cassava cutting not need pollen?' WAIT TIME 10-15 seconds after each probe. Cold-call 4 pairs to describe one change in their model and the evidence for it. Display a rubric for models: includes both pathways; shows cell division in cuttings; shows pollination and double fertilisation; labels ovule to seed and ovary to fruit; includes a barrier to self-pollination; uses at least one Kenyan crop example. Distribute the unit assessment sheet and rubric and set clear rules: individual work, silent, 10 minutes. Remind learners that a pencil and ruler are enough and no blades are required. Exit ticket prompt on the board: 'In two sentences, explain why a lone pawpaw tree can flower but not fruit, and why a cassava cutting can grow into a new plant.' Collect all exit tickets, assessments and CER templates at the door. Close with: 'You have moved from wondering to explaining. Next time you eat githeri, sweet potato or passion fruit, you will know the story behind it.'	tasks apply the ideas to new situations, so learners transfer knowledge beyond the original examples (for example explaining why a sugarcane farmer in Mumias uses setts, or why a tomato farmer relies on pollination).	at least 5 correct labels; double fertilisation shown with two sperm nuclei; a feature hindering self-pollination named; a Kenyan crop used. Unit assessment scored with the rubric (for example 4 levels: beginning, developing, proficient, exceeding). Exit ticket: at least 75% of learners give a correct two-part explanation; those below proficient get follow-up feedback or a small-group re-teach in the next available session.
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D. TEACHER REFLECTION

After the lesson, review the CER samples, models, exit tickets and assessment scores. Which parts of the explanation were strongest (asexual clones, cross-pollination, double fertilisation, fruit development) and which were weakest? Did learners use Kenyan evidence (cassava, maize, hibiscus, passion fruit, pawpaw) accurately? Did the large class size limit participation, and which routines (roles in groups, Think-Pair-Share, station rotation) worked best? Did the cached animations and printed materials work despite limited internet? Were the blade safety rules from Lessons 2 and 5 remembered and followed? Record learners needing extra support and plan feedback or revision, and note changes to improve the unit next year, for example growing cassava cuttings in a school garden in Lesson 1 so learners can watch the growth over time.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe across the unit? Examples: cassava, sugarcane and sweet potato cuttings producing roots and shoots without flowers or seeds; hibiscus, maize, bean and tomato flower parts and how they hinder self-pollination; double fertilisation in the animation; fruit sections of passion fruit and tomato showing seeds attached by placentation; a lone pawpaw or avocado with flowers but no fruit.
What did I learn?	What did we learn? Asexual reproduction (vegetative propagation) forms identical offspring (clones) by mitosis, is fast and needs no partner but offers no variation. Sexual reproduction needs pollination, often cross-pollination, and double fertilisation to form seeds and fruits, and it increases variation. Features such as separate sexes, different timing of anther and stigma maturity and self-incompatibility hinder self-pollination and self-fertilisation. Ovules become seeds and the ovary becomes the fruit.
How does this explain the phenomenon?	How do we explain the phenomenon? Cassava, sugarcane and sweet potato cuttings regrow because the stem tissue divides by mitosis to form roots and shoots, giving a clone with no seeds needed. Maize, beans, pawpaw and avocado must reproduce sexually: a lone flowering tree may not set fruit because its flowers cannot pollinate themselves, so pollen from a partner tree is required for double fertilisation, and only then do the ovules form seeds and the ovary form fruit.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.